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REVIEW

Italian Expert Consensus on the Efficacy and Safety of High Doses of Incobotulinum Toxin A (Bocouture®) in a Full-Face Approach Based on the Electromyographic Analysis

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ABSTRACT

Incobotulinum toxin A (Bocouture®) is a highly purified formulation of botulinum toxin A, devoid of complexing proteins. This characteristic minimises immunogenicity and promotes targeted action. Its high efficacy and safety make it particularly suitable for aesthetic applications, such as full-face and neck rejuvenation. Nonetheless, there exists a lack of standardised dosing protocols for applications in a comprehensive full-face treatment. The present study aims to establish an expert consensus regarding the appropriate dosing and specific injection techniques for incobotulinum toxin A in holistic full-face and neck aesthetic treatments. A multidisciplinary panel comprising six Italian experts in dermatology, plastic surgery, and neurology comprehensively reviewed existing literature (PubMed) on the aesthetic applications of incobotulinum toxin A, supplemented by their clinical experiences. Incobotulinum toxin A was administered to patients with the assistance of a portable electromyography (EMG) device to ensure precise targeting of the muscles. The primary focus was to define optimal injection sites and dosages for various facial and neck muscles, aiming to maximise aesthetic outcomes while minimising toxin diffusion. The unique characteristics of incobotulinum toxin A, including its high purity and low immunogenicity, facilitate its safe utilisation in a full-face approach with a decreased risk of adverse events. The implementation of

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EMG-guided injections significantly enhanced accuracy, thereby improving treatment outcomes by mitigating unwanted toxin dispersal and achieving muscle-specific effects. The consensus reached underscored the validity of a full-face approach, highlighting the necessity of a thorough understanding of muscle anatomy and the application of precise injection techniques. An overwhelming consensus (83.3–100.0%) was attained for most of the proposed strategies, indicating that the full-face treatment regimen employing incobotulinum toxin A yields effective and enduring results.

Keywords: Aesthetics; Bocouture®; Botulinum toxin A; Electromyography; Face and neck aging; Incobotulinum toxin A; Injection site; Neurotoxin; Target muscle; Xeomin®

Key Summary Points

Aesthetic use of botulinum neurotoxin has evolved from treating isolated facial areas to comprehensive full-face approaches, aiming to enhance naturalness, balance, and harmony by considering muscular interconnections.
Incobotulinum toxin A (Bocouture®), owing to its high purity and lack of complexing proteins, is particularly suitable for full-face treatments, minimising immunogenic risk and optimising safety.
Effective full-face treatment requires precise anatomical knowledge and injection technique to ensure accurate delivery into the target muscles, avoiding unintended toxin diffusion.
Currently, there is a lack of standardised guidelines for full-face incobotulinum toxin A administration, prompting the need for evidence-based recommendations.
This consensus review, conducted by six Italian experts in dermatology, aesthetic and plastic surgery, and neurology, defines optimal injection strategies for each facial muscle—including site, depth, dilution, per-site dose, and total dose per muscle—based on literature review and clinical experience.
Injection strategies achieving 83.3–100.0% consensus among panellists were validated using a portable electromyography (EMG) device integrated with the injection syringe. The EMG confirmed correct needle placement by emitting visual and acoustic feedback upon muscle penetration.
Accompanying videos included in the article visually demonstrate the administration techniques and EMG validation process.
Patient-specific anatomy and functionality must guide treatment decisions. The integration of expert-defined criteria with real-time EMG verification enhances injection precision and safety, supporting consistent and effective outcomes.

DIGITAL FEATURES

This article includes digital features, including three videos demonstrating the injection technique at the sites that received the highest level of consensus among the panellists. The videos also show real-time electromyographic feedback confirming the correct positioning and precision of the injections. To view the digital features for this article, please visit https://doi.org/10.6084/m9.figshare.30869543.

INTRODUCTION

The clinical use of botulinum toxin type A (BoNT-A) was first introduced over 50 years ago

for the treatment of strabismus and numerous other therapeutic indications [1–3]. In the early 1990s, the incidental observation of a reversible improvement in glabellar lines [4] initiated the rapid expansion of the cosmetic use of BoNT-A, followed by regulatory approval [2] for the treatment of dynamic upper facial lines [5, 6]. In addition, the off-label use of BoNT-A for age-related wrinkles in the mid and lower face and neck has become widespread, providing safe and effective aesthetic rejuvenation and complete management of facial aging [3, 7–11]. The extensive expertise acquired by clinicians has contributed to refining minimally invasive cosmetic techniques [12], shifting clinical practice from the treatment of isolated facial areas and wrinkle effacement or removal to full-face treatment in a single session [12, 13], aiming to improve facial shaping and provide the patient with natural facial harmony, balance and relaxation [8, 14]. The current goal of botulinum toxin use is to modulate muscle activity and selectively weaken facial and neck muscles to maintain movement control and achieve symmetrical and well-balanced facial recontouring and an expressive facial appearance through a full-face approach [13, 14]. However, this approach requires a thorough knowledge of the anatomical morphology and physiology of face and neck muscles, their interconnections, contraction patterns, as well as the cutaneous insertion of each muscle to ensure better therapeutic outcomes and minimise adverse effects and complications, including unintended adverse effects on contiguous muscles or facial areas [8, 15–18]. Therefore, to establish an individualised treatment plan, a comprehensive patient assessment, including demographic and anatomical factors, is essential [8, 13, 16].

In recent years, injection techniques have advanced significantly, targeting only the muscle to be treated: improper injections into muscles not involved in the treatment can cause abnormal or even antagonistic effects, compromising the expected treatment outcomes [14, 17].

Factors such as injection technique, dosage, volume, and product characteristics may result in toxin spread of the toxin [19] and, in turn, contribute to unwanted events, complications and overall undesirable aesthetic outcomes [18, 20–23].

Guidelines and consensuses published internationally [2, 8, 13, 15, 24–38] remain inconsistent, with dosing recommendations and injection strategies not standardised globally, highlighting differences in ethnic anatomy, preferences for commercially available products and varying clinical approaches.

The appropriate dose of each BoNT-A formulation for individual muscles remains a matter of debate. As a result of their larger and stronger muscle structures, men require a higher dose of product to achieve the same effect as women [3]. However, studies have demonstrated that higher BoNT-A doses are both effective and safe regardless of gender. In a prospective, double-blind, randomised, parallel-group, dose-ranging study of BoNT-A in women with horizontal forehead rhytids, Carruthers et al. observed a greater efficacy and longer duration of response with higher dose of BoNT-A [39]. Further studies have shown that higher doses of BoNT-A do not increase the frequency of adverse events and do not cause a 'frozen' appearance, while still delivering the desired outcome [20, 40]. Furthermore, the administration of higher doses in smaller volumes allows for a greater localization of BoNT-A in the target muscle and a more rapid onset of action [7, 20, 21]. Notably, the injection of BoNT-A formulations free of complexing proteins and with low immunogenicity are ideal for high-dose regimens [38, 41].

Electromyography (EMG) has been widely utilised in neurology to enhance the accuracy and efficacy of botulinum toxin injections when targeting small, deep, or clinically ambiguous muscles, as in the management of focal motor disorders. EMG guidance allows for precise identification of the muscles contributing to abnormal movement or posture, and helps localise the motor point within a muscle, optimising the therapeutic effect and potentially reducing the required dose of botulinum toxin, thereby minimising side effects and antibody formation [42–45].

To the best of our knowledge, this article presents the first Italian expert consensus on the use of incobotulinum toxin A (Bocouture®,

a high-purity product with a very low risk of immunogenicity [41]) in a full-face and neck approach, supported by EMG validation of injection points. This consensus is based on a literature review, the panel's clinical experience, and the results of accompanying electromyographic analyses. The consensus aims to define optimal dosing and injection strategies, optimising aesthetic efficacy while minimising adverse effects. Videos and case studies illustrating injection techniques in conjunction with the use of a portable EMG device are also included.

MATERIALS AND METHODS

A multidisciplinary panel of six Italian experts in the aesthetic use of incobotulinum toxin A (Bocouture, NT 201; Merz Pharmaceuticals GmbH, Frankfurt, Germany), specialised in dermatology, facial plastic surgery, neurology and neurophysiopathology, convened four times to review the available literature on the aesthetic applications of BoNT-A in general, and of incobotulinum toxin A in particular. This review was supplemented by the panellists' clinical experiences with injection procedures. The literature review focused on papers published in international, English-language journals between 1 January 1973 and 31 March 2025 identified through the MEDLINE database (via PubMed). Summaries of the characteristics and prescribing information of various commercially available BoNT-A preparations were also included. The literature search was based on the following terms, combined with the Boolean operators *AND* and *OR*: 'aesthetic medicine/esthetic medicine', 'Bocouture', 'BoNT-A', 'botulinum toxin A', 'BTXA', 'consensus', 'cosmetic practice', 'facial aging', 'facial rejuvenation', 'guidelines', 'incobotulinum toxin A', 'injection technique', 'lifting', 'neuromodulator', 'neurotoxin', 'Xeomin'. Approximately 70 relevant papers were selected and reviewed.

To validate the technical feasibility of the proposed technique and to confirm the accuracy and precision of the injection sites for each targeted muscle, 12 European descent subjects (men and women aged 40–55 years) were enrolled (Table 1). They presented variable facial morphology (e.g. moderate-to-severe lines, Merz scale score 2–3 [46]) and different functional characteristics, reflecting the age group most frequently seeking aesthetic treatment. The cohort included both healthy volunteers and individuals with stable pre-existing medical conditions (e.g. neuromuscular disorders), with a mean age of 49.5 years (2 male, 10 female). Tested muscles were:

- Frontalis and lateral orbicularis oculi in all 12 patients
- Inferior orbicularis oculi in 2 patients
- Corrugator, procerus, and mentalis in 11 patients
- Levator labii superioris alaeque nasi (LLSAN), nasalis, depressor anguli oris (DAO), and platysma in 10 patients

Rather than assessing clinical efficacy or safety, the purpose for this limited cohort was solely to confirm—through EMG feedback—the reproducibility of injection points across subjects with differing anatomical and functional profiles.

During two practical sessions, incobotulinum toxin A was administered using needles connected to a portable EMG device to verify the correspondence between the predefined injection sites and the reaction of the targeted muscles. The electromyographic recording was carried out using a three-channel portable electromyograph (Litebox, Neurosoft, Russia) equipped with needle electrodes specifically designed for botulinum toxin injections (Myobot 30 gauge – 13 mm, Spes Medica Spa, Italy) (Fig. 1a, b) and connected to a laptop. The device provided both visual and audio feedback to confirm the localisation of electrically active muscle areas (Fig. 1c). EMG acquisition settings included free-run mode at 20 ms/division, amplitude 50 µV/division, band-pass filter 10–10,000 Hz, and a sampling frequency of 25,000 Hz, with 60-s trace storage. A valid intramuscular position was defined as an EMG signal exceeding 50 µV in amplitude, associated with the characteristic 'crackling'

Table 1 Characteristics and tested muscles of the 12 enrolled volunteers of European descent (men and women; aged 40–55 years)

Patient no.	Age (years)	Gender	Tested muscles
1	40	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis
2	41	M	Frontalis, lateral orbicularis oculi, corrugator, procerus
3	54	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
4	53	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
5	55	F	Frontalis, lateral orbicularis oculi, inferior orbicularis oculi, LLSAN, mentalis, DAO, platysma
6	43	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
7	55	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
8	52	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
9	50	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
10	55	M	Frontalis, lateral orbicularis oculi, corrugator, procerus, DAO, mentalis, platysma
11	53	F	Frontalis, lateral orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma
12	44	F	Frontalis, lateral orbicularis oculi, inferior orbicularis oculi, corrugator, procerus, LLSAN, nasalis, DAO, mentalis, platysma

DAO depressor anguli oris, *LLSAN* levator labii superioris alaeque nasi

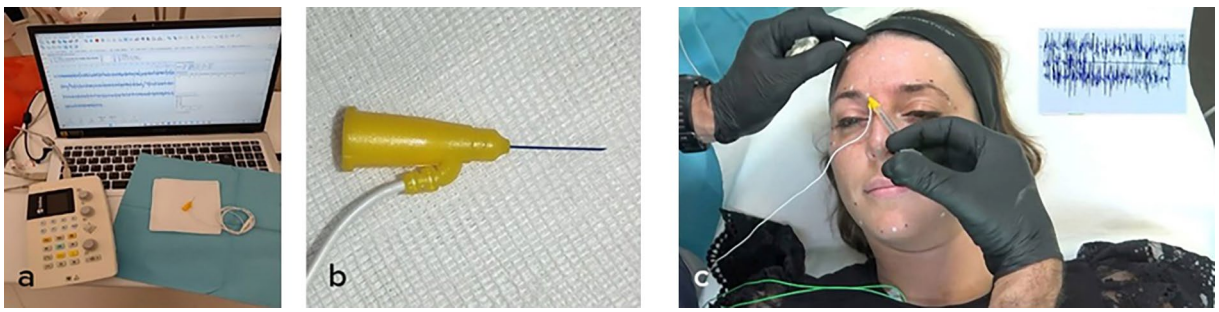


Fig. 1 The electromyograph machine (**a**), needle (**b**), and an example of botulinum toxin injection through the electromyograph machine (**c**)

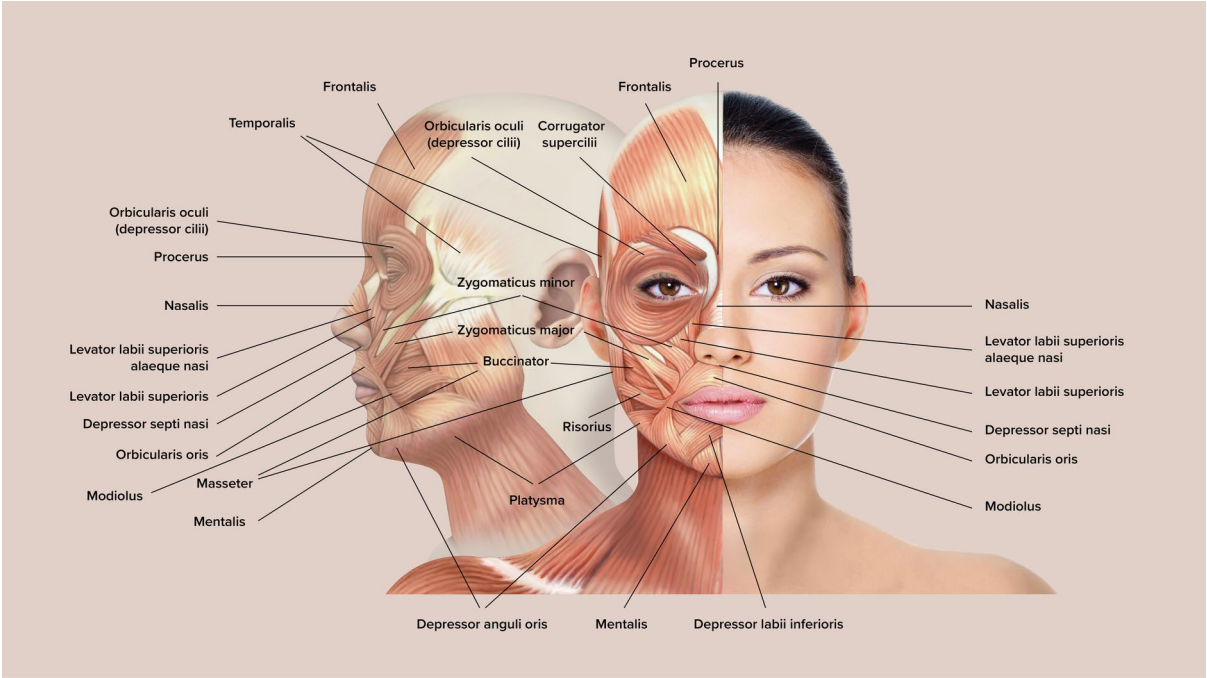


Fig. 2 Representation of facial muscles. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House

sound during voluntary contraction. Instrument-guided injections were administered at anatomical landmarks selected by the expert panel and performed only when the neurophysiologist determined that the EMG signal was optimal. This approach confirmed the needle's location within an active muscle zone before toxin delivery, thereby reducing errors related to injection depth or localisation. Inter-rater reliability was not formally assessed; all EMG-guided procedures were performed by the panel neurophysiologist, ensuring consistent interpretation of the signals.

For clarity, the discussion is organised into four sections: upper face, middle face, lower face, and neck (Fig. 2). Videos and images will accompany the discussion to provide comprehensive documentation of the procedures agreed upon by the panel. Consensus was defined according to the percentage of agreement among the six experts for each proposed injection technique and dosage. A threshold of 83.3% agreement (five out of six panellists) was adopted to indicate consensus. Techniques and dosing strategies endorsed by at least 83.3% of panellists were classified as 'approved'; those supported by less than 83.3% but still considered clinically reasonable were classified as 'plausible'. Techniques and dosing strategies endorsed by ≤ 83.3% but not considered clinically reasonable were classified as 'not approved'.

All patients featured in the images and videos here were treated in accordance with Good Clinical Practice. Ethical committee approval was not required, as the study involved only healthy volunteers undergoing standard, non-therapeutic procedures without experimental interventions, and all participants provided informed consent for the publication of their images and videos, as well as the use of their collected data.

INCOBOTULINUM TOXIN A (BOCOUTURE®)

Properties and Mechanism of Action

Incobotulinum toxin A (Bocouture®) is a highly purified formulation of botulinum neurotoxin

type A, free of complexing proteins [6, 21, 41, 47, 48]. Unlike other available BoNT-A preparations containing accessory clostridial proteins, this purified formulation minimises the risk of inducing neutralising antibodies and secondary non-responsiveness [41, 48–50]. The toxin acts by inhibiting acetylcholine release at the neuromuscular junction, leading to reversible chemodenervation and temporary relaxation of the targeted muscles [6]. These pharmacological properties account for its clinical efficacy in reducing dynamic facial lines [3, 9, 21].

RESULTS

Reconstitution and Dilution

The pharmaceutical form of incobotulinum toxin A is a powder for injection, which must be reconstituted with 9 mg/mL (0.9%) sodium chloride solution for injection before use. A 20–27-gauge (G) short bevel needle is used to introduce the solvent into the vial [3, 6]. After removing the syringe, the vial should be swirled and inverted avoiding vigorous shaking, to ensure proper mixing of BoNT-A with the solvent [6].

Incobotulinum toxin A prescribing information provides eight possible diluent volume options for both 50-U and 100-U vials. For example, adding 1 mL of solvent to a 50-U vial yields a 0.1 mL solution containing 5 U of incobotulinum toxin A [6]. The final dilution volume depends on the personal preference of the practitioners, although the reconstitution range typically falls between 0.25 and 5 mL per 100 U [28, 51, 52].

Table 2 shows the possible dilutions according to the incobotulinum toxin A prescribing information for 50- and 100-U vials.

Table 2 Possible dilutions for incobotulinum toxin A vials (50 and 100 U) as per the product prescribing information [6]

Resulting dose (in U per 0.1 mL)	Solvent added (sodium chloride 9.0 mg/mL (0.9%) solution for injection)	
	Vial with 50 U	Vial with 100 U
5.0 U	1.0 mL	2.0 mL
4.0 U	1.25 mL	2.5 mL

U units

Expert Consensus

Most panellists dilute incobotulinum toxin A for cosmetic applications according to its prescribing information, using a dilution range of 1.0–1.25 mL for 50 U and 2.0–2.5 mL for 100 U. Therefore, this dilution approach achieved a consensus level of ≥ 83.3% and was consequently deemed approved.

In recent years, some panel members have adopted hyperdilutions up to 2 mL for 50 U and 4 mL for 100 U in flat muscles (i.e. frontalis, orbicularis oculi, and platysma), yielding very favourable outcomes and patient satisfaction. This hyperdilution allows for the treatment of a greater number of injection sites with a smaller, less concentrated dose per site. The resulting distribution of the product in the motor plates is slightly broader but controlled, particularly when treating flat muscles. According to the neurologist panellist, who typically treats pathological conditions in flat muscles, higher dilutions increase the number of injection points and help reduce administration errors. When combined with EMG guidance, targeted hyperdilution may enhance treatment efficacy while reducing the total amount of toxin required.

As a result, the consensus rate on hyperdilution does not reach the 83.3%, but panellists consider it plausible.

Results on Injection Procedures and Dosing

In this section, each muscle or muscle group is addressed individually, according to the following systematic approach. First, the relevant mimetic wrinkles and the muscles responsible for their formation are identified. Next, the treatment options for these muscles are discussed, followed by the injection technique

endorsed by the panel. For example, horizontal forehead lines are considered with reference to the contraction of the frontalis muscle, and glabellar lines in relation to the procerus and corrugator supercilii muscles. The techniques proposed by the panellists were based on both their personal knowledge of the literature and their clinical experience, always within the framework of a full-face treatment approach.

In the following section, each muscle or muscle group is described in detail, together with a summary table specifying the type of injection (intradermal, subcutaneous, or intramuscular) and the corresponding depth of administration.

Intradermal therapy involves the introduction of a small volume of drug into the dermal layer of the skin [53]. The dermis is usually between 1.0 and 1.5 mm thick, although its depth varies according to the patient's age and the facial area treated [54–56]. These variations make it difficult to apply a uniform injection protocol. For optimal accuracy, a 4-mm, 30-G needle (or thinner) should be inserted at an angle of about 30° to the skin, allowing the injectate to remain within the dermis while avoiding penetration into the subcutaneous tissue. A small, well-defined papula—a superficial skin elevation—confirms correct intradermal placement [53]. A wider insertion angle of around 45° determines a deeper injection at the level of the dermo-hypodermal junction.

Subcutaneous injection, by contrast, delivers the substance into the adipose and connective tissue located beneath the dermis but above the underlying muscle [57]. The thickness of this layer also changes with age and facial region [58, 59]. This technique typically uses a 13-mm, 30-G needle inserted at roughly a 45° angle to reach the subcutis. Proper injection into this layer produces a small wheal, indicating that the fluid has been correctly deposited in the subcutaneous compartment.

Upper Face

Frontalis Muscle: Forehead Lines

The frontalis, part of the occipitofrontalis muscle, consists of two flat and fan-shaped bellies that include midline fibres which, in some individuals, appear to overlap with the contralateral portions [3, 60–62] (Fig. 2). The muscle originates posteriorly from the galea aponeurotica, superficially aligning with the hairline, and interdigitates anteriorly with the procerus, corrugator, and orbicularis oculi muscles [61, 62]. The frontalis lacks bone attachment and has cutaneous insertions which are responsible for the development of forehead lines [62]. Anatomical and histological variations are frequently observed and are influenced by both individual characteristics and ethnic group differences [3, 16, 62, 63]. To date, four distinct patterns of the frontalis have been identified [63]. These variations can be attributed to individual habits and expressiveness [8].

Contractions of the frontalis result in scalp advancement and brow elevation [3, 60, 64], primarily leading to horizontal forehead rhytids as well as transverse rhytids due to interindividual structural differences [8, 60]. Given that the frontalis attaches to the glabellar complex, which acts as antagonist by depressing the brows, a thorough assessment of the entire upper facial musculature is recommended before neurotoxin injection. This ensures a balanced weakening of the frontalis muscle and appropriate treatment of the overall facial zone [61, 62].

Expert Consensus The injection pattern adopted for improving forehead lines depends on several factors, including the severity of wrinkles, individual facial expression, and the considerable variability in the frontalis muscle's conformation and width, which differs significantly between men and women. Panel members generally avoid injecting the central area of the forehead and maintain a distance of at least 1 cm from the eyebrow arch. Among the panellists, the number of injection sites and units of neurotoxin are typically higher in the upper region of the forehead, with a gradual reduction in dosage towards, and in, the lower region of the forehead (up to 1 U per site).

Each panel member injects the frontalis muscle on both sides at 4–8 points using 5–10 U of incobotulinum toxin A (totalling up to 20 U).

Table 3 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment for the forehead

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Horizontal forehead lines Lift of eyebrow	Frontal belly of the occipitofrontalis	Intramuscular/subcutaneous (standard botulinum) Intradermal (microbotulinum)	4–8 points (standard botulinum) 8–20 points (microbotulinum)	1.25 mL (standard dilution of botulinum) OR > 1.25 mL (higher dilution of microbotulinum)	1.0–2.0 U (standard botulinum) 0.5 U (microbotulinum)	5.0–10.0 U

U units

Depending on the width of the forehead, injections are performed along one or two horizontal or upward oblique rows (Table 3; Fig. 3a, b; Videos 1–3). Although the frontalis muscle is generally larger in men, the panellists tend to use fewer units in male patients to avoid a fixed or frozen effect which is typically undesirable for men. All panellists also perform minimal retreatment of the eyebrow's tail during follow-up session. The panel reached a 100% consensus for this type of treatment of this region (consensus statement: approved). Some panellists suggest the optional use of intradermal injections with the so-called microbotulinum technique targeting the upper portion of the frontalis muscle. This technique involves administering hyperdiluted BoNT-A through multiple injections into the dermis or at the interface between the dermis and the superficial layer of facial muscles. In flat muscles such as the frontalis or platysma, the hyperdiluted BoNT-A diffuses into the underlying muscle, reaching the motor endplates over a wider area [65]. In contrast, intramuscular injection in these muscles can be challenging and may result in overly deep placement, increasing the risk of migration or unintended effects on adjacent, non-target areas.

The number of injection points typically ranges from 8 to 20, with dosage comparable to the standard technique (Fig. 3c). Consensus on this approach remains partial but plausible.

The EMG study was conducted to verify the exact correspondence between the approved injection sites and the reaction of the targeted muscles in frontalis. For this purpose, a 1-mL syringe with a 13-mm, 30-G needle was used. The target muscle was successfully located with a good electromyographic signal, confirmed both visually and acoustically (Videos 1–3).

Glabellar Complex: Glabellar Rhytids

The glabellar complex includes the central procerus muscle, the paired corrugator supercilii muscles and the medial portion of the orbicularis oculi muscle, also known as the depressor cilii [3, 8] (Fig. 2). Together, these muscles act as brow depressors, counteracting the brow elevation caused by the frontalis muscle contractions. The procerus muscle consists of two small

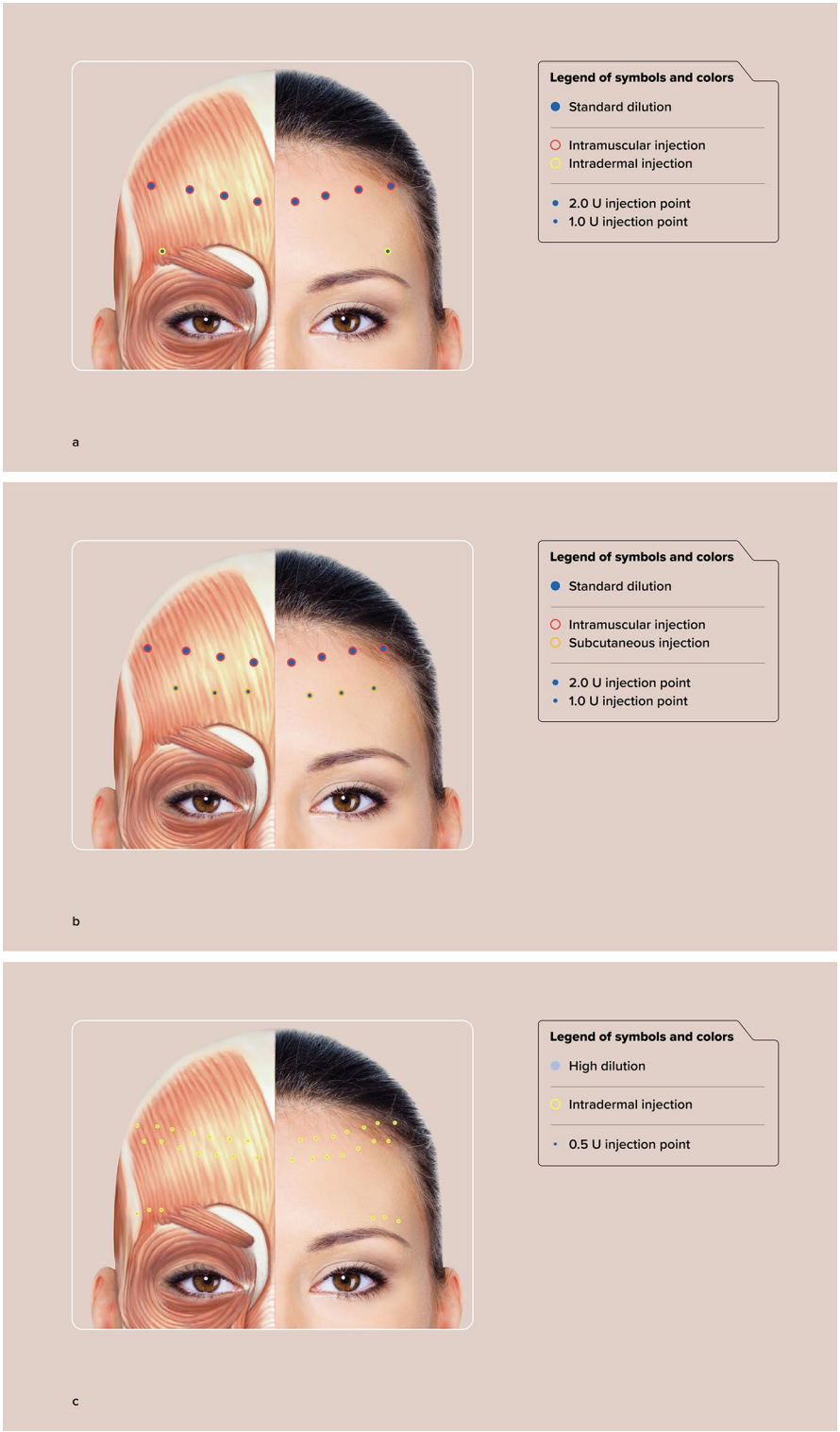


Fig. 3 Injection techniques for the forehead lines with **a** standard botulinum, **b** standard botulinum also including subcutaneous injection points utilizing a lower dose of incobotulinum toxin A compared to (**a**), and **c** microbotulinum. Pictures reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

pyramidal muscles located between and below the eyebrows [63]. It originates from the fascia of the upper nasal region and inserts into the skin between the eyebrows, often blending with the frontalis muscle. Repeated contractions of this muscle over time result in horizontal wrinkles at the nasal root [3, 16]. The paired corrugator supercilii muscles originate from the medial portion of the superciliary arch and insert into the skin above the eyebrows. They pull the eyebrows downward, and hyperactivity in the corrugator supercilii muscles and depressor cilii (partially) is responsible for the formation of vertical glabellar rhytids [3, 60].

All these muscles are targeted during the treatment of glabellar lines with incobotulinum toxin A. When properly injected, the neurotoxin induces a muscular relaxation and chemodenervation, temporarily reducing rhytids in this region [36, 60].

Expert Consensus A preliminary analysis of certain factors should be conducted to guide and individualise the treatment. Panel members recommend evaluating the patient's facial expression both at rest and during muscle contraction. Asking the patient to frown and manually palpating the area can help assess muscle contraction patterns, strength, and muscle mass. The position of the muscle's bony origin is relatively stable, in contrast to its dermal insertion, which exhibits greater variability, particularly in its lateral extension. The lateral extent of the insertion can be assessed by observing the skin depression in the supraciliary zone caused by frowning.

The procerus muscle is typically treated by injecting 2 to 4 U at a single site (Fig. 4a; Videos 2, 3). This is the most common treatment approach among panel members (consensus statement: approved). However, two panel members inject the procerus at two different paramedian points in its lower region (consensus statement: plausible) (Table 4; Fig. 4b).

For each corrugator supercilii muscle, a single toxin injection is typically administered into the corrugator's medial and lateral portions, respectively, using an average of 2.5–4.0 U per injection site. The medial injection is administered deeply and intramuscularly, while the lateral injection is given subcutaneously or intradermally. The panel reached > 83.3% consensus rate on the injection procedure for this muscular region (consensus statement: approved) (Videos 2, 3). Most panellists also administer an additional 1.5–3.0 U along the muscle's dermal insertion (Fig. 4b), extending laterally beyond the pupil line (consensus statement: plausible) and, in some cases, reaching the tail of the eyebrow (Fig. 4c), but this last technique was not approved by the panellists. The injections are performed at an intradermal level (Table 4; Fig. 4). All panellists perform injections at least 1 cm above the orbital rim using a 1-mL syringe with a 13-mm, 30-G needle.

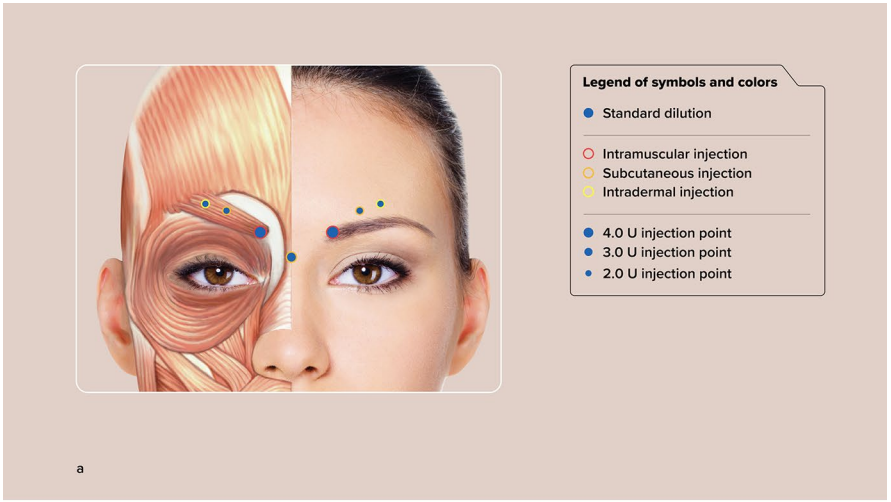
The total toxin dose for the glabellar region ranges from a minimum of 12 U to a maximum of 25 U, with the highest consensus falling between 18 and 22 U.

Electromyographic evaluation, performed on 11 patients, confirmed a strong signal at both the deep medial injection point (bony origin) and the lateral injection point corresponding to the dermal insertion (see Videos 2, 3). No significant signal was detected at the injection sites positioned laterally to the dermal insertion during corrugator supercilii muscle contraction.

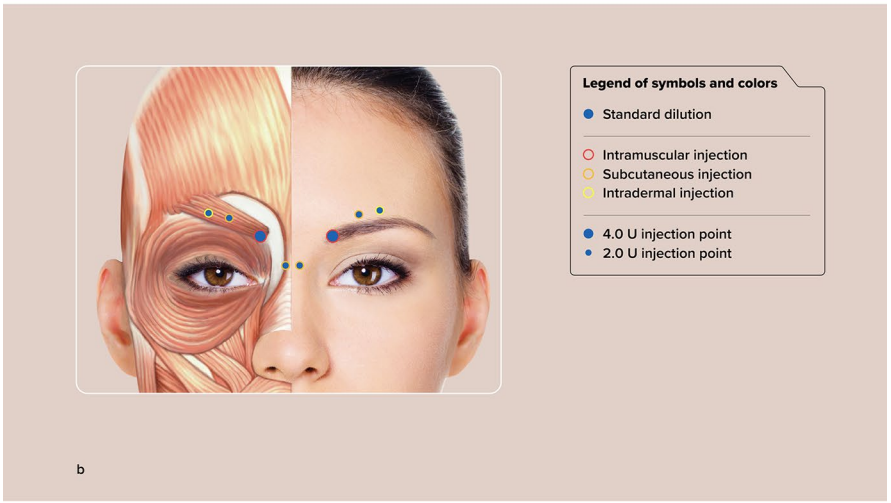
Lateral Portion of the Orbicularis Oculi Muscle: Lateral Canthal Lines or Crow's Feet

The orbicularis oculi originates from the frontal bone, maxilla, medial palpebral ligament, and lacrimal bone, and inserts onto the lateral palpebral raphe as well as the upper and lower tarsal plates. The central fibres of this muscle are very thin, adherent, and inserted into the palpebral skin [63, 66] (Fig. 2). This muscle is divided into three sections: the orbital part, which is the outermost part of the muscle and functions to encircle and protect the orbit; the palpebral part, which enables forceful eye closure during contraction; and the lacrimal part, which contributes to tear drainage [3, 60, 66].

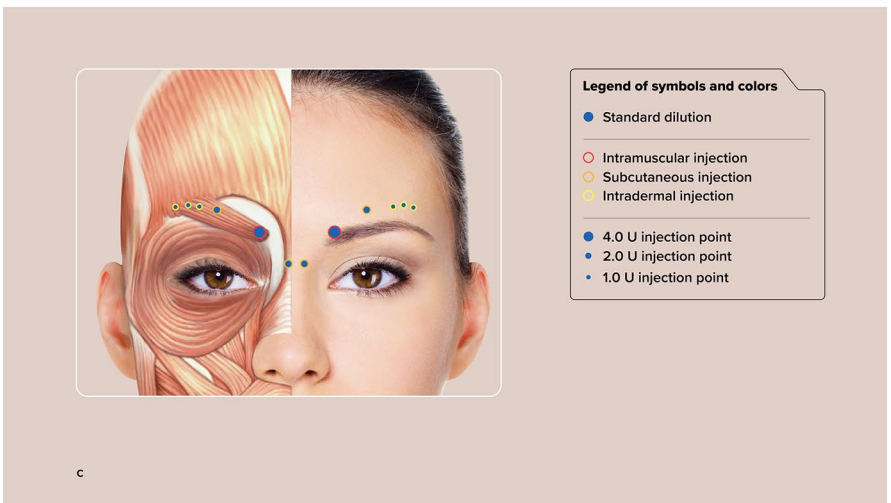
The lateral portion of the orbital part of this muscle is the target for the treatment of lateral canthal lines or crow's feet, which originate


Legend of symbols and colors
Standard dilution
Intramuscular injection
Subcutaneous injection
Intradermal injection
4.0 U injection point
3.0 U injection point
2.0 U injection point

a


Legend of symbols and colors
Standard dilution
Intramuscular injection
Subcutaneous injection
Intradermal injection
4.0 U injection point
2.0 U injection point

b


Legend of symbols and colors
Standard dilution
Intramuscular injection
Subcutaneous injection
Intradermal injection
4.0 U injection point
2.0 U injection point
1.0 U injection point

c

◂**Fig. 4** Injection techniques for the glabellar rhytids with **a** standard botulinum, **b** standard botulinum including two subcutaneous injection points at the procerus, and **c** standard botulinum incorporating two subcutaneous injection points at the procerus along with three additional intradermal injection points per side extending laterally beyond the pupil line. Pictures reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

because of repeated contractions of this muscle section when smiling and squinting [8, 15]. Initially dynamic, these rhytids, radiating from the lateral canthus, can become static over time as a result of aging and solar damage [15].

Expert Consensus As already mentioned, the section of the orbicularis oculi muscle most involved in the development of the lateral canthal lines (or crow's feet) is its lateral portion, where the overlying skin is thin and subtle. This reduced cutaneous thickness requires a very superficial needle insertion. Additionally, this area must be managed carefully to preserve the orbit: for safety reasons, injections should be performed at least 1 cm away from the orbital rim, which is the anatomical landmark, and the injections should be done lateral to the canthal area.

As discussed by the panellists, for each facial side, the number of injections and dosing can vary significantly due to differences in muscle strength, extension, and the potential involvement of the zygomatic muscles in the formation of canthal lines (Fig. 2). Other factors influencing treatment in this area include the severity and depth of the lines, the contraction pattern and muscle activity.

Thus, according to the panel members, the number of injection sites ranges from 2–3 to 5 per side, and most panellists perform 1 or 2 additional injections at the lateral edge of this area (Videos 1–3). Higher doses are typically administered to the upper wrinkles compared to the lower ones. Given the considerable variety of factors involved in the management of canthal lines, incobotulinum toxin A doses range from 1 to 4 U per injection site, and the total amount of injected neurotoxin is 6–14 U per side, with an average of 9–12 U (Table 5; Fig. 5).

As agreed upon by the panellists, two lateral intradermal injections of 2 U each, placed caudally to the tail of the eyebrow, may help achieve a brow lift (Fig. 6).

The tolerability profile, efficacy, and duration of incobotulinum toxin A in this region were demonstrated in a 2015 study by Muti and Harrington [67].

A 1-mL syringe with a 13-mm, 30-G needle is used for this region.

The concomitant treatment of glabellar lines and crow's feet with incobotulinum toxin A results in an eyebrow lift.

The panel agreement on the management of the lateral canthal lines was 83.3%.

Some panellists suggest using the microbotulinum technique for treating the lateral portion of the orbicularis oculi muscle (consensus statement: plausible).

The target muscle was successfully located, as confirmed by consistent visual and acoustic electromyographic signals in 11 tested patients at the main injection sites approved by the expert panel (Videos 1–3).

Inferior Palpebral Portion of the Orbicularis Oculi Muscle: Hypertrophy of Inferior Palpebra

As previously reported, the orbicularis oculi surrounds the upper and lower eyelids. The palpebral portion of this muscle, which is involved in voluntary and involuntary blinking, is further divided into the preseptal and pretarsal sections [66] (Fig. 2; Video 1). Repeated contractions of the inferior pretarsal portion can lead to hypertrophy of the lower eyelid.

Expert Consensus All panellists agree that treating this extremely delicate region requires the superficial insertion of a very thin needle, using very low and hyperdiluted doses of incobotulinum toxin A. A 0.5- or 0.3-mL 30-G or 32-G fixed needle syringe is recommended for this purpose.

Panel members typically inject 0.5 to 1.0 U at each point, performing 1–3 superficial injections

Table 4 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment for the glabella

Indications (aims of treatment)	**Target muscles (muscles involved in the treatment)**	**Preferred injection level (injection technique)**	**Injection points (*n*) per side (injection technique)**	**Dilution (injection technique)**	**Typical incobotulinum toxin A dose per injection point (injection technique)**	**Typical total dose of incobotulinum toxin A (for single muscle)**
Vertical and oblique lines (corrugator) Horizontal line (procerus) Lift of eyebrow's head	Corrugator supercilii muscle Procerus muscle	Intramuscular in the deep bony origin of the corrugator (standard botulinum) Subcutaneous/intradermal in the superficial dermal insertion of the corrugator (standard botulinum) Intramuscular/subcutaneous in the procerus (standard botulinum)	2–5 points in the corrugator (standard botulinum) 1–2 points in the procerus (standard botulinum)	1.25 mL (standard dilution of botulinum)	2.5–4.0 U in the corrugator (standard botulinum) 2.0–4.0 U in the procerus (standard botulinum)	8.0–12.0 U in the corrugator 2.0–4.0 U in the procerus Total dose: 18.0–22.0 U

U units

Table 5 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment for the lateral portion of the orbicularis oculi muscle

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Lateral canthal lines (or crow's feet) Eyebrow's tail lifting	Orbital part of the lateral orbicularis oculi muscle	Intramuscular/ subcutaneous (standard botulinum) Intradermal (microbotulinum)	2–5 points + 1–2 points in the more peripheral part (standard botulinum) OR 3–6 points in the more peripheral part (microbotulinum)	1.25 mL (standard dilution of botulinum) OR > 1.25 mL (higher dilution of microbotulinum)	1.0–4.0 U (standard botulinum) 0.5–1.0 U (microbotulinum)	9.0–12.0 U

U units

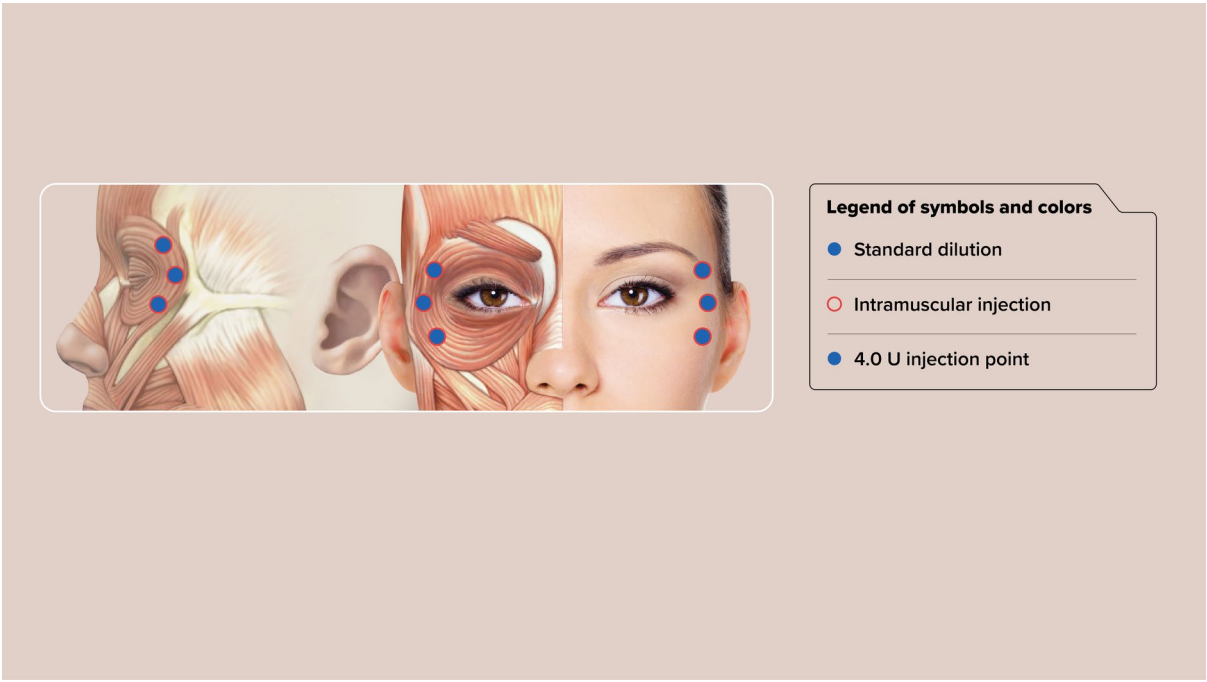


Fig. 5 Injection technique for the lateral canthal lines or crow's feet with standard botulinum. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

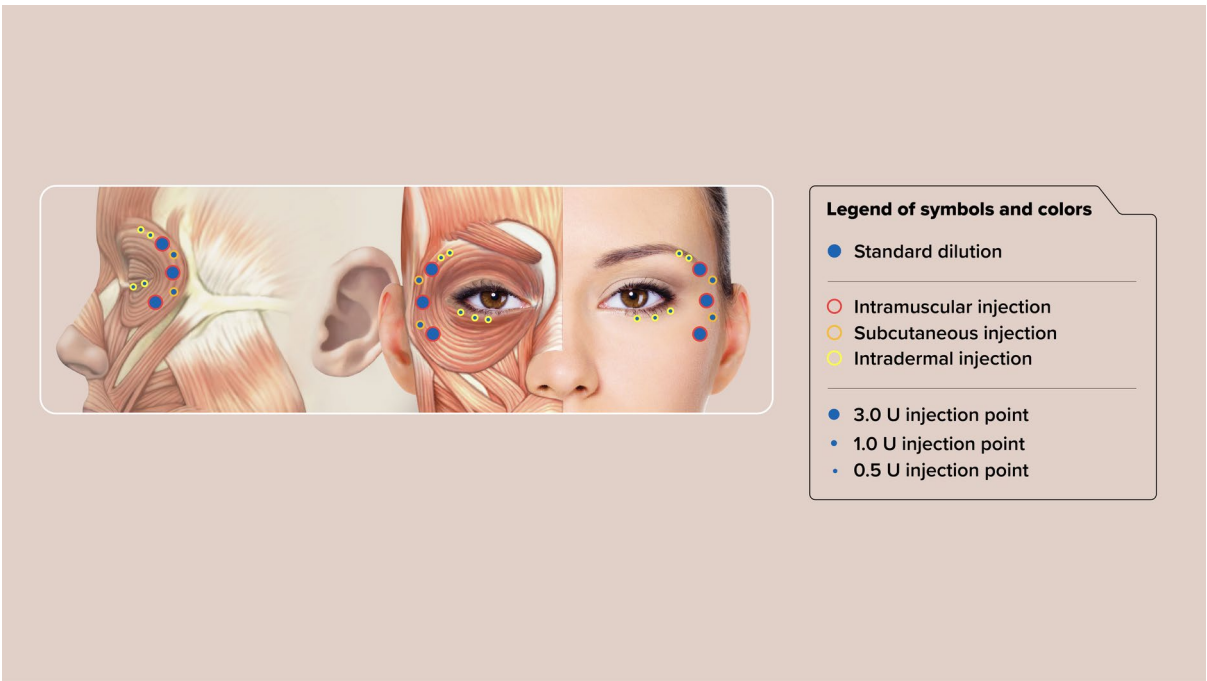


Fig. 6 Injection technique for the combined treatment of lateral canthal lines or crow's feet, browlift, and the inferior palpebral portion with standard botulinum. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

Table 6 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment for the inferior tarsal portion of the orbicularis oculi muscle

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Hypertrophy of inferior palpebra portion of the orbicularis oculi muscle	Inferior tarsal part of the orbicularis oculi muscle	Intradermal (standard botulinum) Intradermal (microbotulinum)	1–3 points (standard botulinum) 2–6 points (microbotulinum)	1.25 mL (standard dilution of botulinum) OR > 1.25 mL (higher dilution of microbotulinum)	0.50–1.0 U (standard botulinum) 0.25–0.50 U (microbotulinum)	1.0–2.0 U
Vertical enlargement of the palpebral aperture						

U units

of 1–2 U of toxin per side. Injection sites should be centrally positioned within the inferior palpebral portion, 1 mm below the tarsal rim or slightly lower, by using a 0.5- or 0.3-mL, 30-G or 32-G fixed needle syringe. The expert panel reached full consensus on this approach. Some panellists suggest using the intradermal microbotulinum technique with double dilution, increasing the number of injection points (2–6), while maintaining the same dose (1–2 U) (Table 6; Fig. 6). Consensus on this approach remains partial and not approved.

The target muscle was identified with a good electromyographic signal, confirmed both visually and acoustically in two patients who have been treated in this area (Video 1).

Middle Face

Nasalis Muscle and LLSAN: 'Bunny Lines'

The nasalis muscle consists of two parts: the alar portion and the transverse portion. The transverse part originates from the maxilla, superolateral to the incisive fossa, and inserts into an aponeurosis across the dorsum of the nose. The alar part arises from the maxilla above the lateral incisor and attaches to the skin overlying the lateral crus of the major alar cartilage. The repeated contraction of the transverse portion is responsible for the development of vertical rhytids [3, 60]. The alar portion functions to pull the posterior part of the columella downward and laterally, whereas the contraction of the transverse portion facilitates the expansion of nares [68]. It results in widening the nares and elongating the nose. When the alar portion of the nasalis is treated with BoNT-A, it can slim the flair of the alar. The LLSAN originates from the frontal process of the maxilla alongside the nose, inserts into the nostril skin, and blends with the fibres of the levator labii superioris (LLS). It is the only muscle which raises the nares and the upper lip, contributing to the formation of the oblique wrinkles seen in bunny lines [68–70] (Fig. 2).

Expert Consensus Both the nasalis muscle and LLSAN contribute to the development of bunny

Table 7 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment of the bunny lines

Indications (aims of treatment)	Target muscles (muscles involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Bunny lines (vertical, oblique) and horizontal rhinion lines	Nasalis muscle LLSAN The lower part of the procerus	Subcutaneous (standard botulinum)	1–3 points (standard botulinum)	1.25 mL (standard dilution of botulinum)	2.0–3.0 U (standard botulinum)	2.0–3.0 U at the nasalis muscle 2.0–3.0 U at the LLSAN 2.0–3.0 U at the lower part of the procerus

LLSAN levator labii superioris alaeque nasi, *U* units

lines and should be targeted for their correction. Depending on the patient's contraction pattern, panel members typically treat the nasalis muscle to correct vertical wrinkles and address oblique lines by injecting the toxin into the LLSAN laterally to the nasal pyramid. These muscles are located just under the skin, so the needle should minimally penetrate the muscle. Bunny lines are sometimes accompanied by more horizontal lines in the rhinion and supra-rhinion areas of the nose. These lines result from the contraction of the lower fibres at the bony origin of the procerus muscle.

The panellists perform one injection centrally into the nasalis muscle and one injection into the central part of the LLSAN muscle on each side. The toxin dose for each injection point is 2–3 U, totalling 8–9 U, administered with a 1-mL syringe with a 13-mm, 30-G needle (Table 7; Fig. 7; Video 3). The lower part of the procerus can be treated by injecting 2–3 U at the subcutaneous level in the radix or rhinion areas of the nose. For the correction of these wrinkles the panellists reached a 100% consensus.

Microbotulinum injection technique may be used by increasing the dilution and injection points number while maintaining the same total units. The consensus statement for this technique was plausible.

The target muscle was identified on the basis of a clear electromyographic signal, which was confirmed both visually and acoustically in 10 patients treated in these muscles (Video 3).

Depressor Septi Nasi: Sagging Nasal Tip

The depressor septi nasi muscle originates from the incisive fossa of the maxilla and inserts into the mobile columella and the alar cartilages of the nose. It contributes to depressing the nasal tip and widening the nostrils [69] (Fig. 2).

Expert Consensus A hypertonic depressor septi nasi may cause a sagging of the nasal tip. This muscle creates ptosis of the nasal tip, which is particularly noticeable when talking or smiling. Most panellists inject the toxin into three sites, with positions that may vary. On average, one or two injection sites are used: one at the base of the columella near the nasal spine, and the other

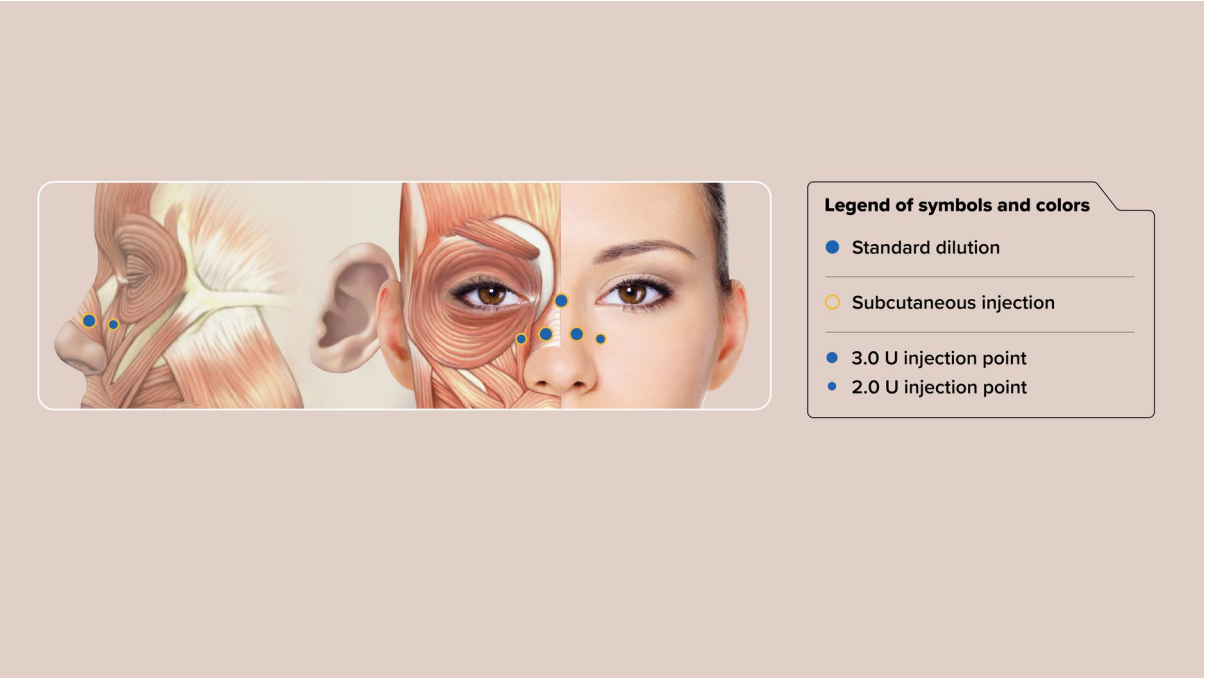


Fig. 7 Injection technique for the treatment of the bunny lines. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

subcutaneously at the centre of the columella. Both injections target the depressor septi nasi muscle, with a total administered dose of 2–6 U of incobotulinum toxin A. A higher dose is used for patients with a more dynamic columella. The other two injection sites are located in the upper part of the alar-facial groove to reduce the contraction of the alar portion of LLSAN, one on each side, each receiving 2 U of neurotoxin per injection, using a 1-mL syringe with a 13-mm, 30-G needle. A contraindication for this specific procedure is upper lip elongation (Table 8; Fig. 8).

For correcting the sagging nasal tip, the panellists reached an 83.3% consensus.

Lower Face

Depressor Septi Nasi and LLSAN: 'Gummy Smile'

A gummy smile is generally caused by the hyperfunction of two muscles: the LLSAN, which lifts and everts the upper lip, and the depressor septi nasi, which draws the nasal tip downward and elevates the central part of the lips. The LLSAN originates from the frontal process of the maxilla and inserts into the superior margin of the lower lateral nasal cartilage, the skin at the alar base, and the skin of the upper lip, where it merges with the orbicularis oris muscle. In severe forms, the LLS and zygomaticus minor muscles (ZMi), which pull the upper lip in a superolateral direction, are also involved [34, 71] (Fig. 2).

Expert Consensus According to the panellists, the LLSAN should be the primary target muscle for treatment to cover the gums when smiling. This muscle is injected with 2.0 U, either intramuscularly or subdermally, laterally to the ala nasi (Video 1). The second target is the depressor septi. Here, the toxin should be injected at a single muscular central point, using a 1-mL syringe, with the needle inserted halfway, at a dose of 2.0–2.5 U. In severe cases, it may be necessary to treat the convergence point of the LLS and ZMi. The injection site, described as the Yonsei point by Hwang and colleagues, is located approximately 1 cm lat-

Table 8 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment of the sagging nasal tip

Indications (aims of treatment)	Target muscles (muscles involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Nasal tip elevation	Depressor septi nasi LLSAN (alaris part)	Subcutaneous (standard botulinum)	1–2 points at the depressor septi nasi (standard botulinum) 1 point at LLSAN (alaris part)	1.25 mL (standard dilution of botulinum)	2.0–6.0 U (standard botulinum) at the depressor septi nasi 2.0 U at LLSAN (alaris part)	6.0–10.0 U

LLSAN levator labii superioris alaeque nasi, *U* units

eral to the ala [72]. A subcutaneous or intramuscular injection of 2.0 U per side is recommended (Table 9; Fig. 9).

In some cases, a follow-up injection after 2 weeks may be necessary, as lateral labial edges may become longer and elevated (the 'Joker effect') as a result of the patient's mimicry. Caution is required when correcting this specific area, as it may not respond as expected or it may be altered by the occurrence of asymmetries.

The target muscle was identified through a distinct electromyographic signal, confirmed both visually and acoustically in 10 patients treated at the approved LLSAN injection site (Video 1).

For the treatment of this region, the panel members reached a 100% consensus.

Orbicularis Oris Muscle: Perioral Lines

Vertical perioral wrinkles have a multifactorial cause, including smoking, photodamage, and expression pattern. The orbicularis oris muscle plays a key role in their generation, with aging further contributing to their worsening [3, 8] (Fig. 2). It originates from the medial aspects of the maxilla and mandible, as well as from adjacent muscles of the cheek, and inserts into the skin and mucous membrane of the lips.

The primary function of the orbicularis oris muscle, which encircles the upper and lower lips, is to protrude the lips and contribute to oral activity and speech [3].

Expert Consensus All the panel members primarily treat the superior lip to reduce the barcode lip lines. Their experience in treating inferior perioral wrinkles is very limited. The goal of the neurotoxin injection in this area is not to augment or increase lip volume.

All panellists inject incobotulinum toxin A very superficially at four sites, two on each hemilip. The injection sites are located along the vermilion border of the lips. Each site receives 1 U of incobotulinum toxin A, leading to a total dose of 4 U administered with a 1-mL syringe with 13-mm, 30-G needle (Table 10; Fig. 10). A tangential needle insertion is recommended to ensure an intradermal injection. In this region,

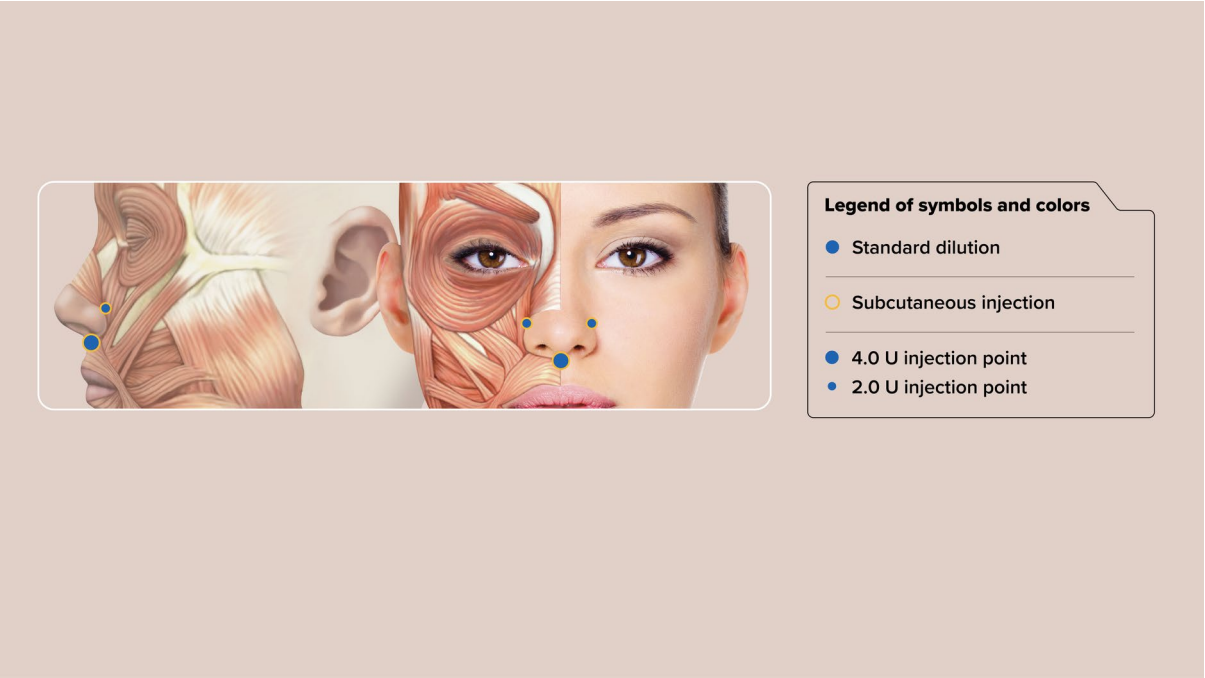


Fig. 8 Injection technique for the treatment of the sagging nasal tip. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

the effect of the neurotoxin is very short-lived, requiring a re-treatment after 2 or 3 months.

The panel members reached a 100% consensus on the treatment of this region.

Masseter and Temporalis Muscles: Hypertrophy

Originating from the zygomatic bone and zygomatic process and inserting distally on the lateral surface and angle of the mandibular ramus, the powerful quadrangular masseter is one of the primary muscles responsible for mastication. Its contraction contributes to mandible elevation and protrusion, working with the pterygoid and temporalis muscles [3, 73] (Fig. 2).

Masseteric hypertrophy, a benign enlargement of the masseter muscles, leads to a square jawline shape, commonly observed in Asian individuals and less frequently in Western populations, where genetic and habitual factors may contribute to its occurrence [3, 26, 34, 73].

Expert Consensus As a result of the robust mass and strength of these muscles, massive doses of incobotulinum toxin A are typically applied by the panellists, although aesthetic management of these muscles is not frequent. Treatment aims generally at reducing excessive mass (as frequently requested by Asian patients) and to address functional disorders, including bruxism, malocclusion, or oromandibular dystonia. Some panellists reported never having treated the temporal muscle, whereas the neurologist of the panel typically treats both muscles concomitantly.

On average, 3–5 injection points per side are used, and 25–50 U of neurotoxin are deeply injected into each side of the masseter muscle (leading to a total incobotulinum toxin A dose of 50–100 U), using a 1-mL syringe with a 13-mm, 30-G needle (Fig. 11a, b). As noted by one panellist, the deep and the superficial heads of the masseter should be treated performing a retrograde injection.

The panel members reached an 83.3% consensus for the aesthetic management of this region.

Table 9 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment of the gummy smile

Indications (aims of treatment)	Target muscles (muscles involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Excessive gingival display	LLSAN Depressor septi nasi LLS ZMi	Subcutaneous or intramuscular (standard botulinum)	1 point at the LLSAN Second target: 1 point at the depressor septi nasi In severe cases, 1 convergence point (Yonsei point) of the LLS and ZMi (standard botulinum)	1.25 mL (standard dilution of botulinum)	2.0 U (standard botulinum) at the LLSAN, depressor septi nasi, LLS, and ZMi	6.0–10.0 U

LLS levator labii superioris, *LLSAN* levator labii superioris alaeque nasi, *U* units, *ZMi* zygomaticus minor

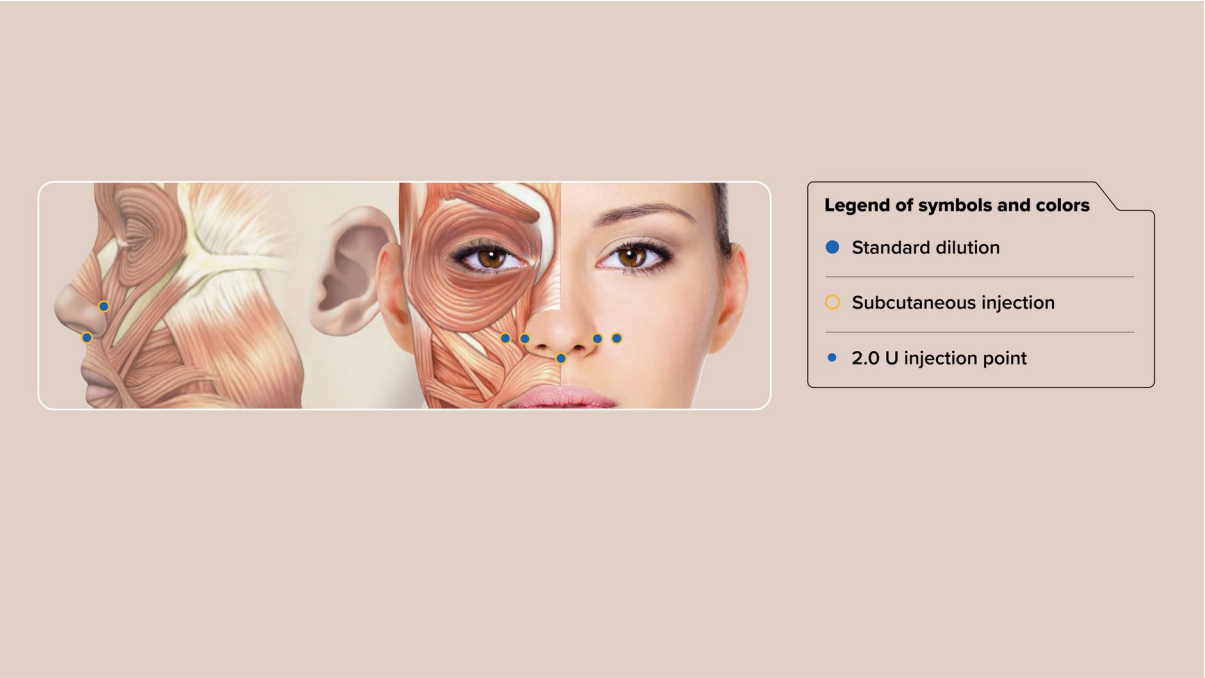


Fig. 9 Injection technique for the treatment of the gummy smile. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

Mentalis Muscles: Dimpled Chin or Peau d'Orange Appearance

The mentalis muscle originates from the incisive fossa of the mandible and inserts into the skin of the chin. The contraction of the paired mentalis muscles is mainly responsible for everting the lower lip and elevating the chin. These muscles interdigitate and merge with the fibres of the orbicularis oris and the depressor labii inferioris [3, 60] (Fig. 2). Over time, hyperfunction of these muscles can contribute to the development of a deep transverse crease and, by flattening the chin, the development of a peau d'orange appearance of the chin's skin surface [8, 60]. Loss of collagen and subcutaneous fat in the chin may exacerbate these conditions [26]. Relaxation of the mentalis muscles through neurotoxin injections can smooth the cutaneous surface [36].

Expert Consensus Among the panellists, the preferred injection technique is to treat the mentalis muscles at two bilateral and symmetric sites, with 3 to 6 U of incobotulinum toxin A injected at each point, totalling 6–12 U into both the muscle bellies. Full needle penetration is used (Table 11; Fig. 12; Video 3). One panellist performs a single median injection of 8 U, allowing the toxin to diffuse to the muscles. Another panellist suggests the use of one central injection point when treating the DAO as well. The placement of injection sites may depend on the prominence of the chin. Caution is advised during needle insertion to avoid incorrect involvement of the depressor labii inferioris muscle or the orbicularis oris which could cause asymmetry or lip dysfunction.

In certain cases, this area may be re-treated after 2 weeks to optimize the response of motor endplates that remain active by injecting 2 U of incobotulinum toxin A more superficially.

A 1-mL syringe with a 13-mm, 30-G needle is used in this region.

The panellists reached an 83.3% consensus on the correction of the wrinkled chin, with specific reference to the two lateral injection sites.

The target muscle was identified with a good electromyographic signal, confirmed both

Table 10 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment for the orbicularis oris muscle

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Perioral lines	Orbicularis oris	Intradermal or subcutaneous (standard botulinum)	2 points on each hemilip (standard botulinum)	1.25 mL (standard dilution of botulinum)	1.0 U (standard botulinum)	4.0 U

U units

visually and acoustically in 11 patients treated at the two approved lateral injection sites (Video 3).

DAO: 'Marionette Lines'

The fan-shaped DAO muscles, which depresses the corner of the mouth, originate with a broad base along the mandibular body, anterior to the masseter, and insert into the oral commissures at the modiolus complex [3, 74].

Vertical lines at the oral commissures, commonly referred to as marionette lines or melomental rhytides, can develop as a result of DAO hyperactivity which inverts and depresses the corners of the mouth, flattening the lips and resulting in a sad or disapproving expression [3, 26, 60] (Fig. 2).

Chemodenervation of the DAO can elevate the oral commissures [3, 60].

Expert Consensus All panellists, except one (who does not use botulinum toxin A at all for the treatment of marionette lines, relying exclusively on dermal fillers), treat this area by injecting 2–4 U in a single site on each side at the subcutaneous level, with a 1-mL syringe with a 13-mm, 30-G needle. The panel members' discussion primarily focused on injection site placement. While all members agree to place the needle 0.5 cm lateral to the marionette lines, half of the panel suggests injecting halfway between the corner of the mouth and the jawline, while the other half recommends an injection point 1 cm above the jawline in a lower position (Table 12; Fig. 13a, b; Videos 1, 3). This divergence reflects the anatomical complexity of this muscle due to its overlap with the depressor labii inferioris. Panellists highlight the difficulty in recognizing the cause of marionette lines, which may result from either ptosis of the cheek soft tissues or DAO overtone. They state that only in patients with DAO overtone can incobotulinum toxin A provide good results. Panellists also note that the DAO and mentalis muscles are often treated simultaneously because of their synergistic action and frequent concurrent hyperactivity.

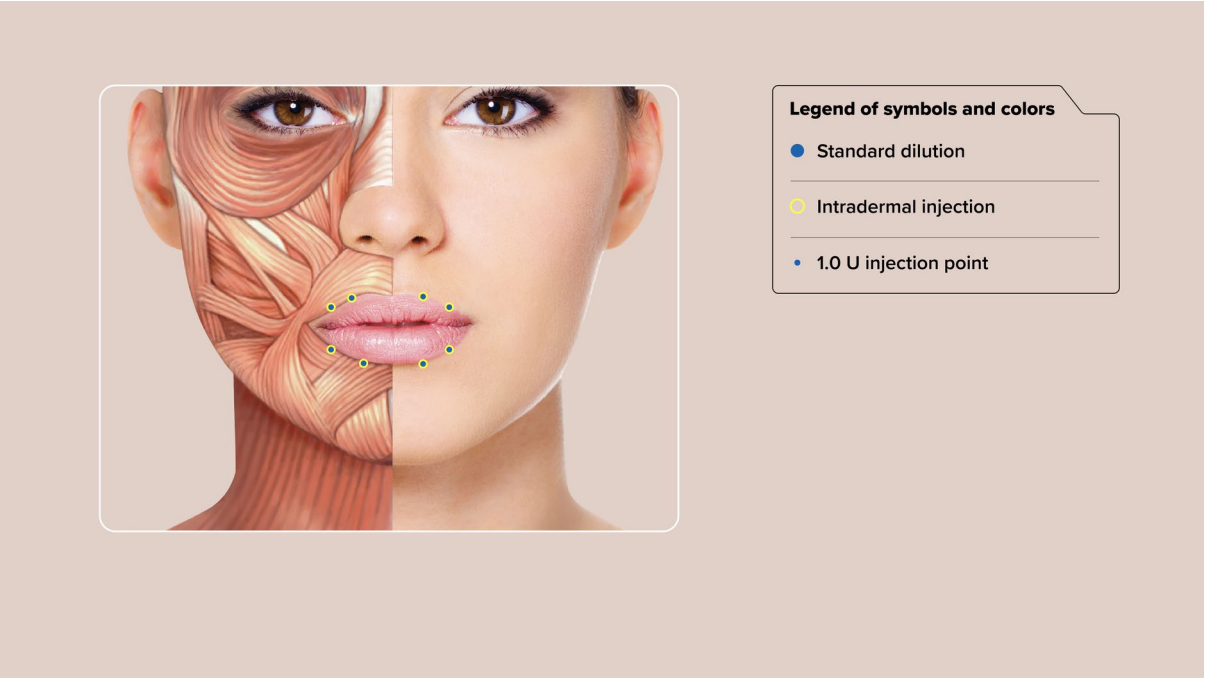


Fig. 10 Injection technique for the treatment of the perioral lines. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, unit

The DAO can be identified with high degree of certainty using ultrasound imaging to guide treatment. Electromyographic signal localization of the DAO muscle confirms findings from functional anatomy studies and enables botulin toxin injection in the lateral part. This technique reduces the risk of inadvertent injection into the DLI (depressor labii inferioris or quadratus labii inferioris) muscle which could cause lip asymmetry. One panel member observed that upper face toxin treatment often triggers hypertone of the DAO and mentalis muscles, necessitating their treatment in the same session.

The panel reached approximately 50% consensus on the procedure for 'marionette line' correction. The consensus statement was considered plausible for both the injection sites.

The target muscle was identified by a clear electromyographic signal, confirmed visually and acoustically both at the midpoint between the corner of the mouth and the jawline, as well as at the injection site located 1 cm above the jawline (Videos 1 and 3).

Neck

Platysma: Vertical Platysmal Bands, Lower Face Soft Tissue Ptosis, and Slight Neck Laxity

The broad, thin, and superficial platysma muscle originates in the pectoral and deltoid muscles, extending over the clavicle along both sides of the neck and under the skin near the mandible [3, 8]. It inserts onto the medial part of the lower border of the mandible, as well as into the skin of the lower lip, oral commissure, and cheek. It is a powerful depressor of the soft tissue of the lower face and mandible, also contributing to the downward pull of the oral commissures [3, 8, 26, 34, 60] (Fig. 2).

The position and number of platysmal bands of the neck are variable. These bands become more prominent with aging, often accompanied by horizontal necklace lines and fat deposits which can worsen the aesthetic appearance. The treatment of hyperactive platysmal bands is particularly effective in younger patients where skin elasticity is still preserved

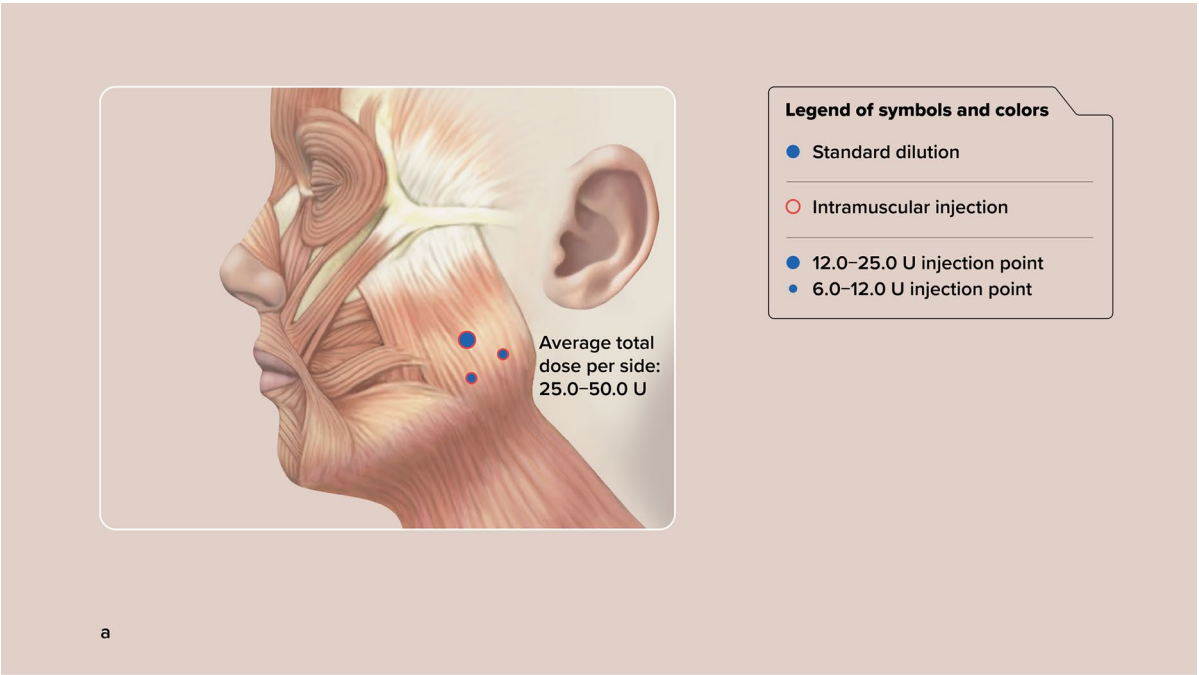


Fig. 11 Injection technique for the treatment of the hypertrophy of the masseter and temporalis muscles performed with either 3 (**a**) or 5 (**b**) injection points. Pictures reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

Table 11 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment for the mentalis muscle (dimpled chin or peau d'orange appearance)

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Chin peau d'orange Horizontal labiomental fold	Mentalis muscle	Intramuscular	1 point per side (standard botulinum) OR 1 central point (standard botulinum)	1.25 mL (standard dilution of botulinum)	3.0–6.0 U (standard botulinum)	6.0–12.0 U

U units

and gravitational sagging is not yet significant [8, 26, 36].

Expert Consensus Firstly, it is worth noting that the lower face should be treated comprehensively in a single session to achieve complete muscular rebalancing of the entire face using incobotulinum toxin A. Therefore, treatment of the platysma should always be combined with neurotoxin treatment of the mentalis and DAO.

According to the panellists, the number of injection sites per platysmal band varies between 7 and 8 points, with injection sites spaced 1.5 cm apart along the bands, typically totalling 4–6 points per band. The average neurotoxin dose recommended by the panellists is 1.5–2.5 per injection site, totalling 10.5–15.0 U per platysmal band. One panellist gradually decreases the neurotoxin dose from 2.0 U at the top to 1.5–1.0 U at the bottom. The needle should penetrate very superficially, almost intradermally or subcutaneous, to avoid impairing neck muscles and preventing adverse events, such as dysphagia or dysphonia (Table 13; Fig. 14a, b). A tangential needle insertion is recommended to better control the injection depth.

In the neck region, the panellists perform the Nefertiti lift to recontour the jaw and neck profiles. For this procedure, 1.25 U (for hyperdiluted doses) to 3 U of incobotulinum toxin A are injected at each of the three injection points along the jawline using a 1-mL syringe with a 13-mm, 30-G needle. There is variability in injection placement among the panellists: maintaining the jaw bone as a landmark and injecting following the mandibular shape, one panellist injects the neurotoxin above the mandibular arch, another panellist injects on the bone, and the others prefer to inject below the mandibular arch, always maintaining superficial placement. One panellist suggests the microbotulinum technique, involving several intradermal microinjections of double-diluted incobotulinum toxin A in an area starting from the lower part of the cheek and extending along the neck corresponding to the platysma's extension. The average dose is 25 U per side. This technique is more effective for soft tissue lifting of the lower face but less effective for reducing platysmal bands [75].

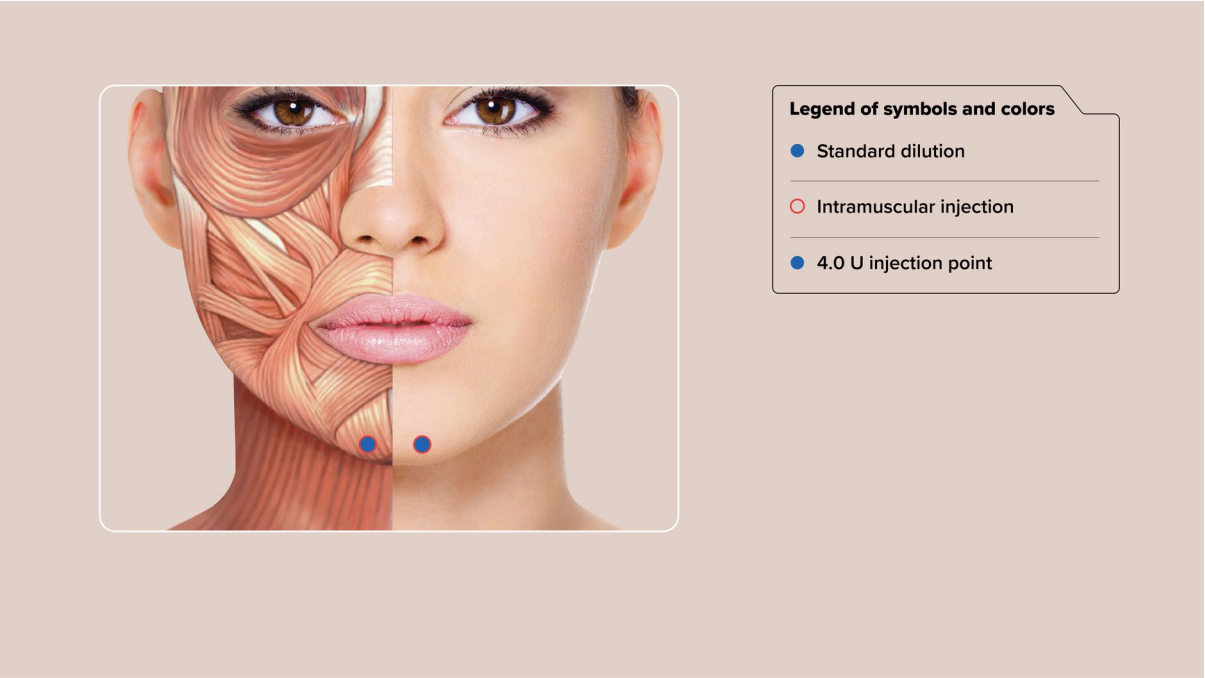


Fig. 12 Injection technique for the treatment of the dimpled chin or peau d'orange appearance. Picture reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

For the treatment of the platysmal bands and jawline, as well as for the platysma treatment with microbotulinum, the panel members reached a 50% consensus, indicating that both the techniques were considered plausible.

The target muscle was identified through a good electromyographic signal, confirmed both visually and acoustically along the platysmal bands in ten patients treated in these injection sites (Video 1).

Some clinicians utilize incobotulinum toxin A to treat the décolleté, aiming to reduce the contraction of the lower portion of the platysma muscle near its insertion points with the pectoralis fascia. The use of incobotulinum toxin A in this area reduces décolleté wrinkles by minimising lower platysma contraction. However, most panellists do not perform this treatment because of its limited efficacy and the high dosage required.

An atlas-style summary of the techniques endorsed by the panellists is shown in Table 14.

Electromyographic Verification

Electromyographic verification of a good electrical signal was performed at the injection points that received the highest consensus among the panellists. During the consensus meetings, 12 patients underwent EMG verification to ensure accurate needle positioning and precision of the injection points. As accuracy and outcomes depend largely on the ability to selectively activate the target muscles, all subjects were clinically examined beforehand and instructed on how to perform the appropriate facial movements. Electromyographic feedback for three of these patients is presented in the included videos (Videos 1–3). The needle electrode was inserted at the identified points until the electromyograph device recorded a strong signal of muscular electrical activity during muscle contraction. The injector administered the incobotulinum toxin A only after confirming a positive electromyographic signal. No injections were performed in its absence. All

Table 12 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment of the marionette lines

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
DAO overactivity Drooping mouth corners Marionette lines	DAO	Subcutaneous	1 point per side (standard botulinum)	1.25 mL (standard dilution of botulinum)	2.0–4.0 U (standard botulinum)	4.0–8.0 U

DAO depressor anguli oris, *U* units

the tested injection points demonstrated a strong signal, confirming the presence of muscle fibres and therefore the correct position and precision of the injection.

Full-Face Approach

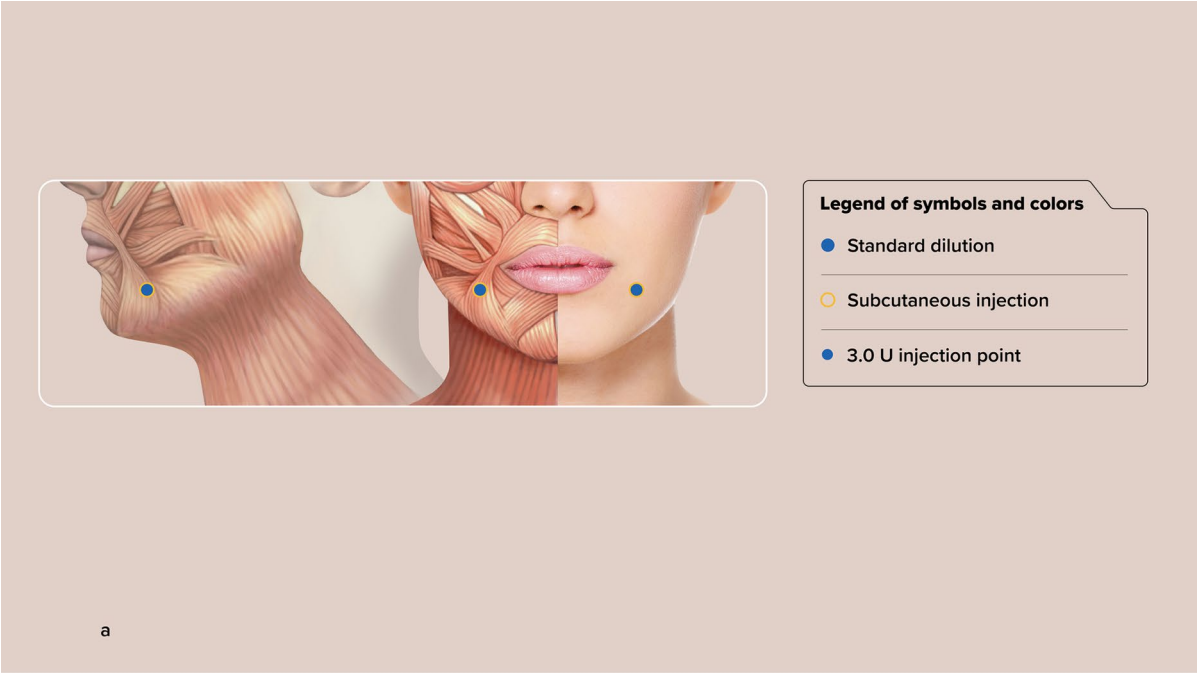
All panellists (100% consensus) routinely treat the upper face muscles with incobotulinum toxin A in a single session. One panellist frequently combines the treatment of both the upper and lower faces in the same session. They observed that by inhibiting muscular contractions in both regions simultaneously, the zygomaticus muscles in the mid-face become slightly hyperactive. This results in a comprehensive softening, smoothing, and balancing of the entire face, leading to high patient satisfaction. The remaining panellists agree with this method, but highlight that the treatment of mentalis muscle, depressor septi nasi, and upper face in a single session is often dictated by the patient's individual needs and financial resources.

DISCUSSION

To our knowledge, this is the first Italian expert consensus on the aesthetic use of incobotulinum toxin A in a full-face and neck approach, supported by EMG validation of injection sites. Previous studies have confirmed the efficacy and safety of BoNT-A in the upper face, but protocols remain heterogeneous, and no standardised recommendations have been available until now. This consensus therefore contributes practical guidance to harmonise clinical practice.

Rationale for a Full-Face Approach with BoNT-A

Facial rejuvenation with BoNT-A should be approached comprehensively, treating the face and neck as a whole rather than isolated muscles [12, 49], since interventions in one region may



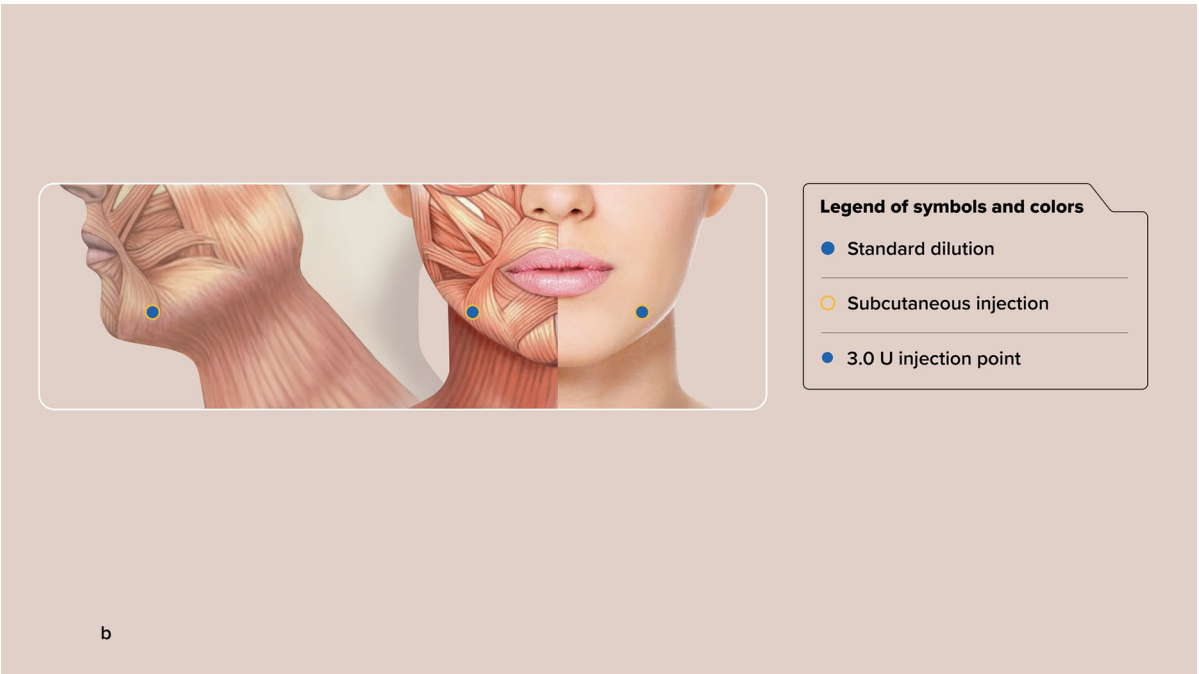


Fig. 13 Injection technique for the treatment of depressor anguli oris (DAO) either by injecting halfway between the corner of the mouth and the jawline (**a**) or by placing an injection point 1 cm above the jawline in a lower position (**b**). Pictures reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

Table 13 Consensus recommendations and expert panel insights on incobotulinum toxin A treatment of the platysma muscle

Indications (aims of treatment)	Target muscle (muscle involved in the treatment)	Preferred injection level (injection technique)	Injection points (*n*) per side (injection technique)	Dilution (injection technique)	Typical incobotulinum toxin A dose per injection point (injection technique)	Typical total dose of incobotulinum toxin A (for single muscle)
Platysmal bands Lower face soft tissue ptosis Slight neck laxity	Platysma muscle	Intradermal or subcutaneous	4–6 points for each platysma band 8–12 points per side (standard botulinum) 40–60 points per side (microbotulinum)	1.25 mL (standard dilution of botulinum) OR > 1.25 mL (higher dilution of microbotulinum)	1.5–2.5 U (standard botulinum) OR 0.4–0.6 U (microbotulinum)	25.0–40.0 U per side Total dose: 50.0–80.0 U

U units

affect adjacent areas through muscular interconnections and physiological interactions [8, 12]. This full-face strategy aims to restore balance and harmony by temporarily reducing hyperdynamic muscle activity, improving contour, shape, and symmetry with a favourable safety profile [10, 12, 14, 49]. Full-face protocols, involving the simultaneous treatment of upper, mid, and lower facial areas, generally require higher cumulative doses compared with conventional upper-face treatments. This concept is in line with previously published clinical experience [76].

To optimise outcomes, the panellists recommend highly purified formulations with low immunogenicity, such as Bocouture®, which offers well-documented efficacy, purity, and tolerability [6, 11, 41]. Its absence of complexing proteins limits toxin spread and supports the safe use of full-face regimens [20, 49, 50].

Clinical Implications and Safety

Randomised controlled trials (RCTs) have consistently confirmed the efficacy and safety of incobotulinum toxin A. A phase III randomised, double-blind, placebo-controlled trial by Kerscher et al. demonstrated efficacy across the glabellar, forehead, and lateral canthal regions with rapid onset, long duration, and significant improvements in multiple upper-face areas and a favourable safety profile lasting up to 120 days [10]. Similarly, Polacco et al. showed that higher doses (60–100 U) produced greater efficacy and longer-lasting effects than conventional regimens without additional adverse events or a 'frozen' appearance [20]. Overall, adverse events are uncommon, usually mild to moderate, and mainly include headache and transient eyelid ptosis [11, 49]. In our consensus, no adverse events were reported, except for a single case in which muscles previously affected by paralysis, and later partially reinnervated with pathological synkinesis, were treated. Despite the low dose administered and EMG guidance, the injection produced a temporary 'overtreatment', which resolved spontaneously over time. Importantly, no neutralising antibodies have been reported in patients treated with incobotulinum toxin A

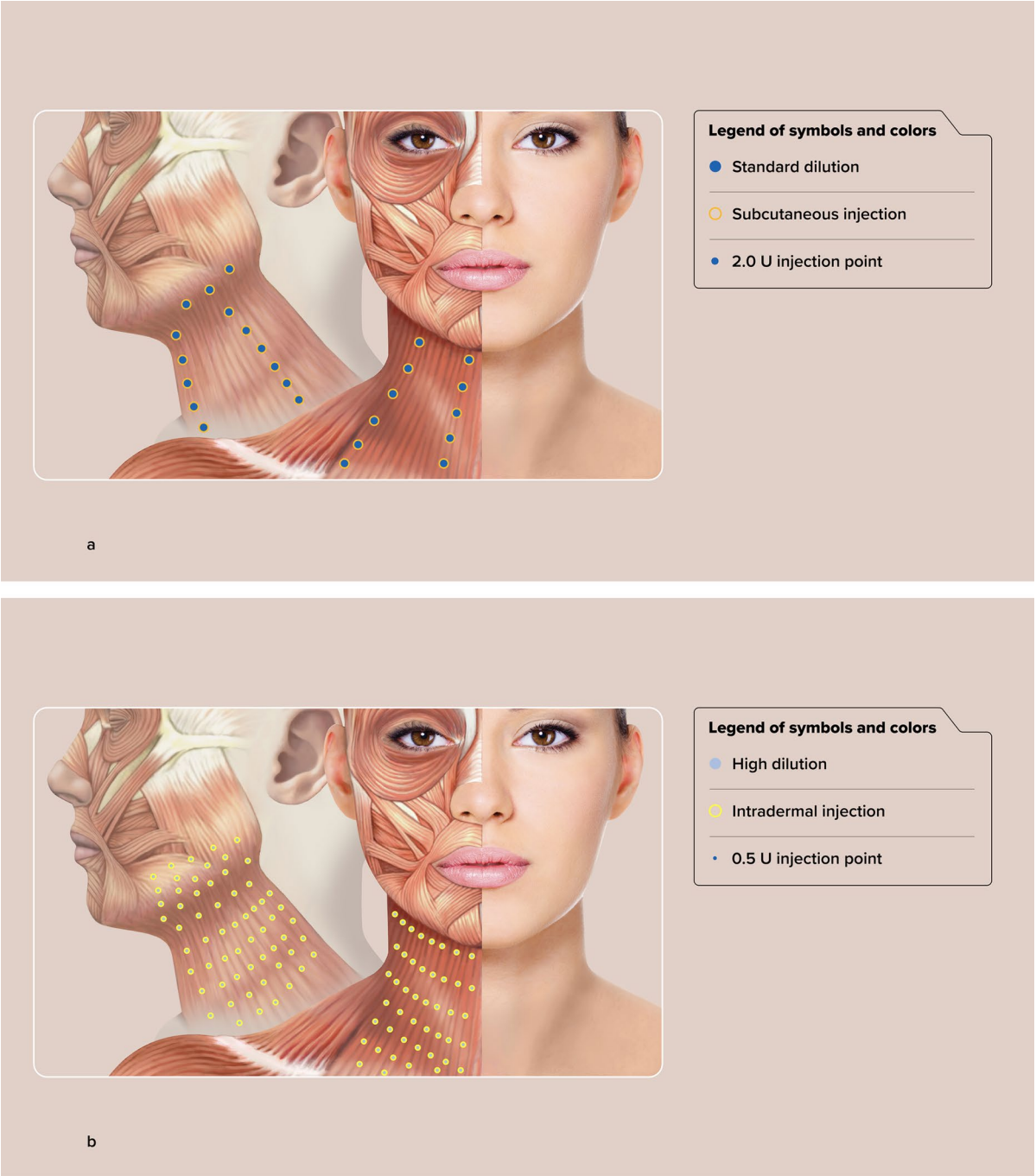


Fig. 14 Injection technique for the treatment of the vertical platysmal bands (**a**), lower face soft tissue ptosis and slight neck laxity (**b**). Pictures reproduced from *XVII Anatomical Tables. Injection Points of Botulinum Toxin*. Author: P Fundarò. Courtesy of OEO Firenze Publishing House. U, units

[49], and the favourable safety profile has also been confirmed in neurological patients receiving high doses for therapeutic indications [77]. These results are consistent with international consensus papers and reviews recommending highly purified, low-immunogenic formulations

Table 14 Summary of consensus-based injection techniques for incobotulinum toxin A in the full-face approach

Muscle/area	Clinical signs/wrinkles	Units (range, per side)	Injection techniques and points	Consensus level
Upper face				
Frontalis (frontal belly of the occipitofrontalis)	Horizontal forehead lines Lift of eyebrow	5–10 U per side	Standard botulinum (intramuscular/intradermal; 5 points per side: Fig. 3a) Standard botulinum (intramuscular/subcutaneous at lower dose; 7 points per side: Fig. 3b) Microbotulinum (intradermal, 17 points per side: Fig. 3c)	Approved ≥ 83.3% Approved ≥ 83.3% Plausible < 83.3%
Glabellar complex (corrugator supercilii, procerus)	Vertical and oblique lines (corrugator) Horizontal line (procerus)	Corrugator supercilii: 5–8 U per side Procerus: 2–4 U per side	Standard botulinum (intramuscular/intradermal/subcutaneous; 2–3 points per side: Fig. 4a) Standard botulinum (intramuscular/intradermal/subcutaneous; 1 central point or 2 paramedian points: Fig. 4a,b)	Approved ≥ 83.3% Plausible 83.3%
Orbicularis oculi (lateral)	Crow's feet	6–14 U per side	Standard botulinum (intramuscular; 2–5 points per side: Fig. 5) Microbotulinum (see text)	Approved ≥ 83.3% Plausible < 83.3%

Table 14 continued

Muscle/area	Clinical signs/wrinkles	Units (range, per side)	Injection techniques and points	Consensus level
Orbicularis oculi (medial/inferior palpebra)	Hypertrophy of inferior palpebra	2–4 U per side	Standard botulinum (intradermal; 3 points per side: Fig. 6) Microbotulinum (intradermal; 2–6 points per side: see text)	Approved ≥ 83.3% Plausible < 83.3%
Middle face				
Nasalis muscle and levator labii superioris alaeque nasalis	Bunny lines	8–9 U total	Standard botulinum (subcutaneous; 5 points total: Fig. 7) Microbotulinum (see text)	Approved ≥ 83.3% Plausible < 83.3%
Depressor septi nasi	Sagging nasal tip	2–6 total 2–4 U per side	Standard botulinum (subcutaneous; 3 points total: Fig. 8) 1 point per side	Approved ≥ 83.3% Approved ≥ 83.3%
Lower face				
Depressor septi nasi and levator labii superioris alaeque nasi	Gummy smile	8–10 U total	Standard botulinum (subcutaneous; 1 point midline, 2 points per side: Fig. 9)	Approved ≥ 83.3%
Orbicularis oris	Perioral lines	4 U total	Standard botulinum (intradermal; 4 points per side, including 2 points on each hemilip: Fig. 10)	Approved ≥ 83.3%
Masseter (temporalis muscle not assessed due to limited panellist experience)	Jawline contour, hypertrophy	25–50 U per side	Standard botulinum (intramuscular; 3 points per side: Fig. 11a) Standard botulinum (intramuscular; 5 points per side: Fig. 11b)	Approved ≥ 83.3% Approved ≥ 83.3%

Table 14 continued

Muscle/area	Clinical signs/wrinkles	Units (range, per side)	Injection techniques and points	Consensus level
Mentalis	Dimpled chin	6–12 U total	Standard botulinum (intramuscular; 2 points per side: Fig. 12) Standard botulinum (intramuscular; 1 point midline: see text)	Approved ≥ 83.3% Plausible < 83.3%
Depressor anguli oris (DAO)	Marionette lines	2–4 U per side	Standard botulinum (subcutaneous; 2 points per side, each one halfway between the corner of the mouth and the jawline: Fig. 13a) Standard botulinum (subcutaneous; 2 points per side, each placed 1 cm above the jawline in a lower position: Fig. 13b)	Plausible < 83.3% Plausible < 83.3%
Neck				
Platysma	Vertical platysmal bands, lower face soft tissue ptosis, and slight neck laxity	20–40 U total	Standard botulinum (subcutaneous; 8–12 points per side: Fig. 14a) Microbotulinum (intradermal; 40–60 points per side: Fig. 14b)	Plausible < 83.3% Plausible < 83.3%

Consensus levels were defined as: 'approved' ≥ 83.3% (endorsed by the majority of panellists) or 'plausible' < 83.3% (not reaching the threshold but considered clinically reasonable). Techniques outside these categories were not included in the final consensus

U units

such as Bocouture® for full-face aesthetic treatments [38, 41].

As a result of complex interactions between facial muscles, spread of BoNT-A into non-targeted muscles remains a clinical concern [8, 12, 21]. These risks are influenced by dosage, injection technique, and anatomical variability [16]. The limited spread of incobotulinum toxin A contributes to more precise and localised delivery [49], while EMG guidance further enhances accuracy by confirming correct needle placement and active muscle targeting.

Dilution strategies also have safety implications. Hyperdiluted incobotulinum toxin A has

been used effectively in delicate areas such as the inferior palpebra, orbicularis oculi, frontalis, and platysma, achieving high patient satisfaction with no evidence of increased diffusion. A prospective open-label study in 15 women with moderate-to-severe upper facial lines (glabellar, forehead, and lateral canthal lines) used a dilution of 100 U in 7.5 mL, yielding a total dose of 35 U per patient, and demonstrated effective improvement of wrinkles with no adverse events related to increased toxin spread [52]. A 2020 Italian retrospective real-life study applied a double-dilution protocol (100 U in 5 mL), followed by gentle massage of the injected area, in 197 patients. This approach reduced the required amount of BoNT-A while maintaining the efficacy of conventional therapy, with no serious adverse events and high patient satisfaction [78]. Finally, another Italian randomised, double-blind, two-arm study compared two dilutions of incobotulinum toxin A (31.25 vs. 90.91 U/mL) in 40 women with moderate-to-severe lateral canthal lines. Both regimens achieved comparable therapeutic efficacy, although the higher dilution produced smoother and more homogeneous results [79]. Consensus panellists noted that dilution should be adapted to muscle morphology: standard dilutions for non-laminar muscles, higher dilutions for laminar muscles such as the frontalis, orbicularis oculi, and platysma [65, 80]. However, high-volume dilution is discouraged in smaller mimetic muscles because of the risk of excessive spread and unwanted effects [81–87]. For larger muscles, by contrast, higher dilution may represent a safe and effective strategy to enhance diffusion across evenly distributed motor endplates.

Main Outcomes of the Consensus

This consensus highlights the impact of anatomical variability and patient-specific factors on treatment outcomes. Variability in muscle thickness, morphology, and contraction patterns, as well as factors such as age, sex, and facial expressivity, influence both dosing and injection depth. These individual features are best addressed through a tailored full-face approach. The panel achieved strong consensus on injection patterns and dosing for most regions of the upper (e.g. frontalis and inferior portion of the orbicularis oculi), middle and lower face (e.g. LLSAN, mentalis, depressor septi nasi, and orbicularis oris). By contrast, only partial consensus was reached on optional approaches (i.e. the use of microbotulinum techniques) and the treatment of DAO and platysma muscle. Safety issues were raised: superficial injection techniques in delicate regions, careful dosing and EMG guidance were recommended to reduce the risk of diffusion and associated complications such as dysphagia, dysphonia, and asymmetry. These recommendations reflect consensuses, pharmacodynamic studies and safety analyses already available in the literature [8, 18–23, 26, 88]. Comparing the incobotulinum toxin A dosing and injection patterns reported by the panellists with the most recent Italian consensus report (where onabotulinum toxin A was used) demonstrated that the neurotoxin dosage suggested by the panellists herein is frequently considerably higher, whilst injection techniques are fairly similar [36]. No published guidelines or consensuses describe similar BoNT-A dosing and number of injections. Although differences are not substantial, full correspondence is rare [2, 8, 13, 15, 24–38]. Similarly, the incobotulinum toxin A dosage, the injection sites and procedures described herein partially diverge from recommendations or indications reported in international guidelines, whereas higher similarity can be recognized when comparing this consensus to the other Italian ones. The panel also highlighted that combining BoNT-A with hyaluronic acid fillers may enhance facial rejuvenation, particularly in specific regions. de Maio et al. proposed this strategy for the lower face, with filler providing support before neurotoxin injection [15], while the panel suggested it may also benefit certain upper-face areas. The 2020 HARMONY study further supported this panfacial approach, showing that combined treatments achieved not only aesthetic improvements but also favourable psychological effects [89]. However, most RCTs on BoNT-A have focused on single-region interventions (e.g. glabellar,

Table 15 Comparative summary of international consensus recommendations on BoNT-A dosing and injection sites

Muscle/area	Raspaldo et al., 2010–2011 [25, 26] (onabotulinum toxin A)			Sundaram et al., 2016 [90] Pan-Asian (incobotulinum toxin A)			Sundaram et al., 2016 [13] Global (onabotulinum toxin A)			Signorini et al., 2021 [37] (onabotulinum toxin A)		
	IP (*n*)	Dose/IP (U)	Total dose (U)	IP (*n*)	Dose/IP (U)	Total dose (U)	IP (*n*)	Dose/IP (U)	Total dose (U)	IP (*n*)	Dose/IP (U)	Total dose (U)
Frontalis	3–5	1–2 U	F: 10–20 M: 10–30	6 × row 2 rows	0.1–5	2–32	4–8	2–4	8–25	4–5/side	2–4	20
Glabella	2–5	4–5	Higher than 25	3	2–4	6–25	3–7	2–4	12–40	5–7	6	30–50
Crow's feet	3–4/side	2–4	6–16/side	3–4	2–4	4–16/side	1–5/side	1–4	6–15	3/side	6	36
Orbicularis oculi (medial/inferior palpebra)	1–2/side	1–2	2–4	2–3/side	0.5–1	1–2	1/side	0.5–1	0.5–1/side	1	0.5–1/side	0.5–1/side
Nasalis muscle and levator labii superioris alaeque nasalis (gummy smile)	1/side	1–2/side	2–4	1/side	1–2	1–2	1–2/side	0.5–2	1–4	2/side	2	8
Depressor septi nasi (bunny lines)	1/side	2–4	4–8	1/side	0.5–5	3–4	2–3	2–4	4–8	1/side	2–3	4–6

Table 15 continued

Muscle/area	Raspaldo et al., 2010–2011 [25, 26] (onabotulinum toxin A)			Sundaram et al., 2016 [90] Pan-Asian (incobotulinum toxin A)			Sundaram et al., 2016 [13] Global (onabotulinum toxin A)			Signorini et al., 2021 [37] (onabotulinum toxin A)		
	IP (*n*)	Dose/IP (U)	Total dose (U)	IP (*n*)	Dose/IP (U)	Total dose (U)	IP (*n*)	Dose/IP (U)	Total dose (U)	IP (*n*)	Dose/IP (U)	Total dose (U)
Orbicularis oris (perioral lines)	Up: 4 Lo: 2–4	Up to 1	Up: 4 Lo: 2–4	2–6	0.5–1	1–8	2–5	0.5–1	1–5	Up: 2	0.5–1	2–4
Masseter	1–5/side	–	18–30/side	3–5/side	4–6	20–40	1–5/side	5–15	15–40	3–5/side	5–22	25–50/side
Mentalis	2/side	3–5/side	6–10/side	1–2	2–4	2–16	1–4/side	2–3	4–10/side	1–2	4–10	8–10
Depressor anguli oris (DAO) (marionette lines)	1/DAO	1–2/side	2/side	1	2–4	2–4	1–2	2	2–4	1	2–4	4–8
Platysma	2–4/band	2	–	3–20	2–4	–	3–6/band	1–3	6–12/band	5/band	2	40

The values reported here can be compared with those presented in Table 14, which summarise the consensus achieved by the present expert panel

U unit, *IP* injection point, *n* number, *F* female, *M* male, *Up* upper lip, *Lo* lower lip

forehead, and lateral canthal lines [10, 11, 49]) or on dose-ranging studies in relatively narrow populations [20]. Hence, they may not fully reflect the complexity of real-world aesthetic practice. In contrast, consensus recommendations—particularly the present work—address practical aspects that are rarely captured in RCTs, such as injection depth, dilution strategies, full-face approaches, and the use of EMG guidance. In this context, EMG guidance is particularly valuable, as it reduces the uncertainty associated with surface landmarks: when the needle is not within the targeted muscle, no signal is detected, thus preventing unintended injection into non-target structures. This real-time functional feedback enhances the reproducibility of injections across diverse patient profiles, supporting its translation into daily clinical practice.

Our consensus findings are broadly consistent with previously published international recommendations; while the treatment of the frontalis, glabellar, and lateral canthal lines is supported by strong and relatively uniform consensus, important differences remain for several muscles, particularly in the middle and lower face. These differences reflect not only anatomical and functional variability but also cultural and practice-related preferences across regions, underlining the need for adaptable treatment strategies. Table 15 summarises the recommendations reported in international consensus statements, highlighting both convergences and heterogeneities in upper- and lower-face, and neck protocols.

Our consensus is not intended to replace the methodological strength of RCTs, but rather to complement trial evidence by filling knowledge gaps on full-face protocols, multiregional treatment strategies, and long-term safety. The integration of RCT data with expert consensus and other international recommendations therefore provides a more comprehensive framework to guide the safe and effective use of incobotulinum toxin A in aesthetic medicine.

Limitations

This consensus has several limitations. It reflects the practice of a limited number of Italian experts and does not directly compare incobotulinum toxin A with other formulations. In addition, the lack of long-term follow-up precludes conclusions on durability. The variability of international approaches, particularly across ethnic groups, further limits the generalisability of these recommendations. These limitations confirm the complementary—rather than substitutive—role of consensus alongside controlled trials.

CONCLUSIONS

The indications presented here for incobotulinum toxin A treatment of the face and neck are based on the panellists' personal clinical experience and the validation of injection techniques using EMG in a limited cohort. The review of published trials and international consensus statements has proved the efficacy and safety of incobotulinum toxin A for aesthetic use. However, these indications are specific to incobotulinum toxin A, should not be extrapolated to other toxin formulations, and are aimed at standardising clinical practice.

Key points include:

- *Individualised assessment and treatment planning*: Treatment should be tailored to each patient and to the specific toxin used since each formulation has a unique pharmacological action.
- *Treatment for hyperactivity of mimetic muscles*: Incobotulinum toxin A is an adequate treatment for hyperactivity of the mimetic muscles, which may determine facial disharmony and aging enhancement due to an imbalance in tone of the muscular groups across different facial areas. A full-face approach is often required to restore balance and harmonise facial muscular activity. When combined with hyaluronic acid fillers, it can further enhance facial rejuvenation outcomes.

- *Dilution strategies*: The choice of dilution should reflect the anatomical and functional characteristics of the targeted muscle. Higher dilution may be used for flat and larger muscles (platysma, frontalis, and orbicularis oculi), while a standard dilution is recommended for smaller muscles.
- *Injection patterns*: Injection patterns should be adapted to individual anatomical and functional characteristics. This consensus has identified and verified standard patterns for each muscle group through EMG, offering a reliable reference for injection planning.
- *Injection depth*: The depth of injection plays a crucial role in determining the effect of incobotulinum toxin A. Intramuscular injections yield greater neuromodulation and stronger blockade of motor endplates, whereas subcutaneous or intradermal injections reduce neuromodulation and minimise complications from unintended effects on non-target muscles, especially in anatomically complex areas.
- *Instrumented guidance*: Diagnostic tools, such as EMG, may improve injection accuracy in specific muscle area, particularly those prone to adverse effects or suboptimal outcomes. Future technological developments may lead to easier-to-use devices, making precise botulinum toxin administration accessible to a broader range of practitioners.

These recommendations are based on expert opinion supported by EMG validation in a small, single-country cohort, and should be interpreted in this context. Broader, multicentre, and international studies are warranted to confirm these findings, refine dilution strategies, and further evaluate the role of instrumented guidance in routine aesthetic practice.

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Declarations

Conflict of Interest. Salvatore Piero Fundarò has received consultancy and speaker honoraria from Evolus Pharma B.V., IBSA Italy, Merz Aesthetics Italia Srl, and Relife Srl. Alessandro Lozza has received honoraria for scientific consultancy and training activities from Merz Aesthetics Italia Srl and Spes Medica Spa. Massimo Renzi and Nicola Zerbinati has received scientific consultancy honoraria from Merz Aesthetics Italia Srl. Gabriele Ferruccio Muti and Stefano Santoro declare that they have no competing interests.

Ethical Approval. All procedures involving healthy volunteers were conducted in accordance with Good Clinical Practice guidelines. Written informed consent was obtained from all individuals for participation, data collection, and publication of images and videos.

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